

GRADE 12 IT PAT Phase 2 2020



CLIFTON

Scholarship

Leadership

Sportsmanship

Community

Figure 1 (School, 2017)



Figure 2 (Coetzee, 2018)

home affairs

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

By: Byron Kraus

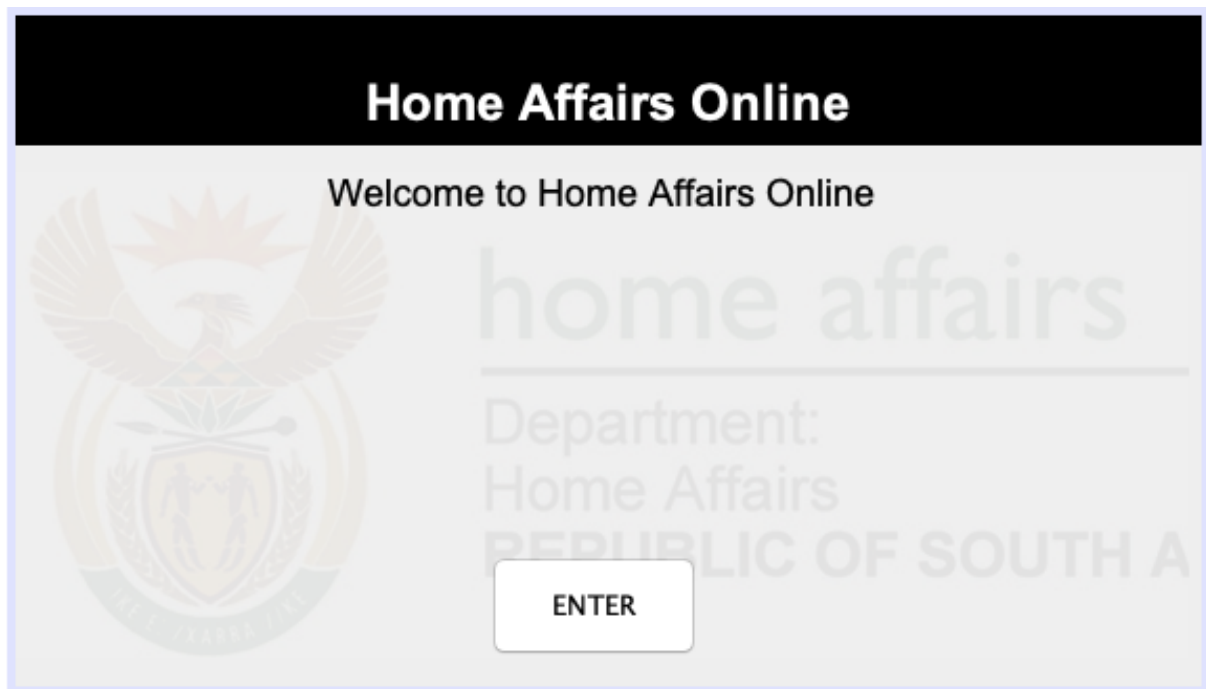
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1. Applicant Interface Design

1.1 Welcome Screen



Input	Enter button – this may be clicked to take the applicant to the login screen.
Output	The welcome screen will vanish and the login screen will appear. Messages will appear in the output box to show that the program has connected to the server.
Processing	The program will connect to the database, but will not process any data.
Data Storage	No data is being stored on this screen and the program will try connect to the database.
Security Group	All applicants, including the applicants who do not have log in details, can access this screen.
Error Checking	No error checking will occur on this screen.

1.2 Login Screen

Home Affairs Online

Enter your login details:

ID number

Password

Incorrect ID or password. Please try again

Home Affairs Online

Enter your login details:

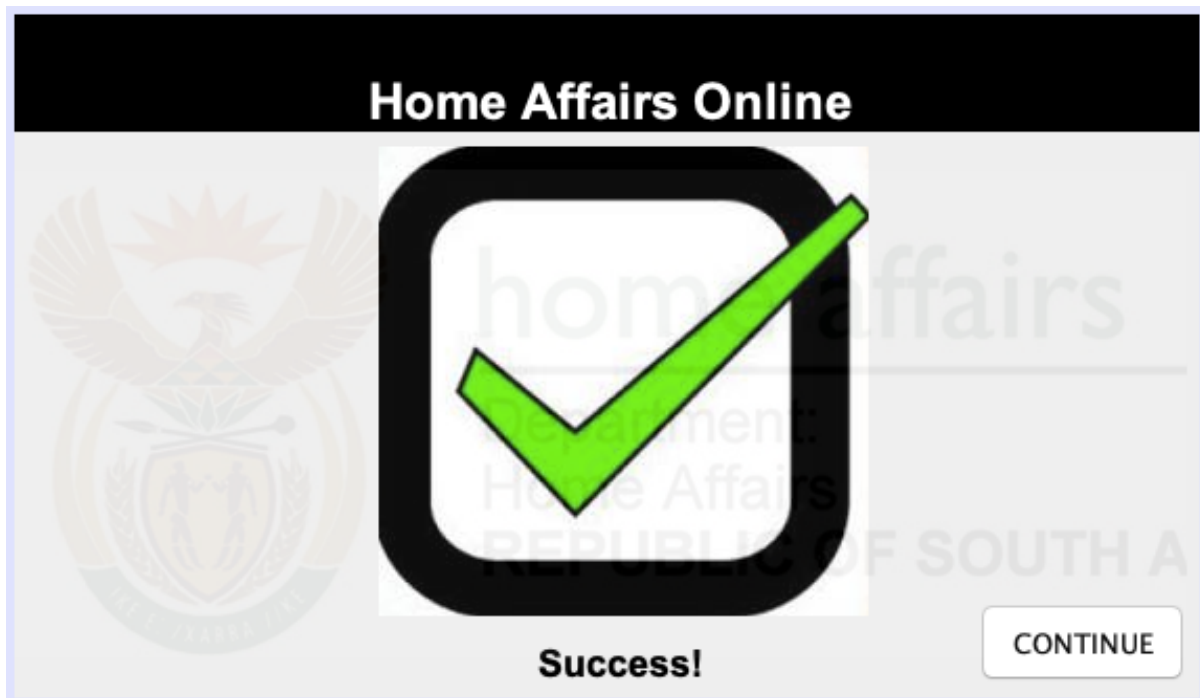
ID number

Password

Input	Text fields – The applicant may enter their ID and password into the respective fields, the program will automatically cover the password with * symbols. Login button – This button will allow the applicant to check if the information entered by them is correct.
Output	Once the login button is pressed, the login screen will disappear. • If the combination is not in the text file – a warning will be shown on the login screen in red, shown in the second screen. • If the combination is in the text file – the user of the program will be taken to either the applicant home screen or the staff member home screen depending on whether they are a staff member or not.
Processing	This program will look through a text file containing ID and password of all users to determine whether the login details match any in the text file. • If the login is successful – The program will work out whether the applicant is a staff member or an applicant. It will determine whether it must go to the applicant home screen or the staff home screen. • If the login is unsuccessful – the program will close the text file and make the screen show the warning label.
Data Storage	The applicant's ID, password and whether they are a customer or a staff member is stored in a text file. The program will search through this file to try locate the matching applicant name and password.

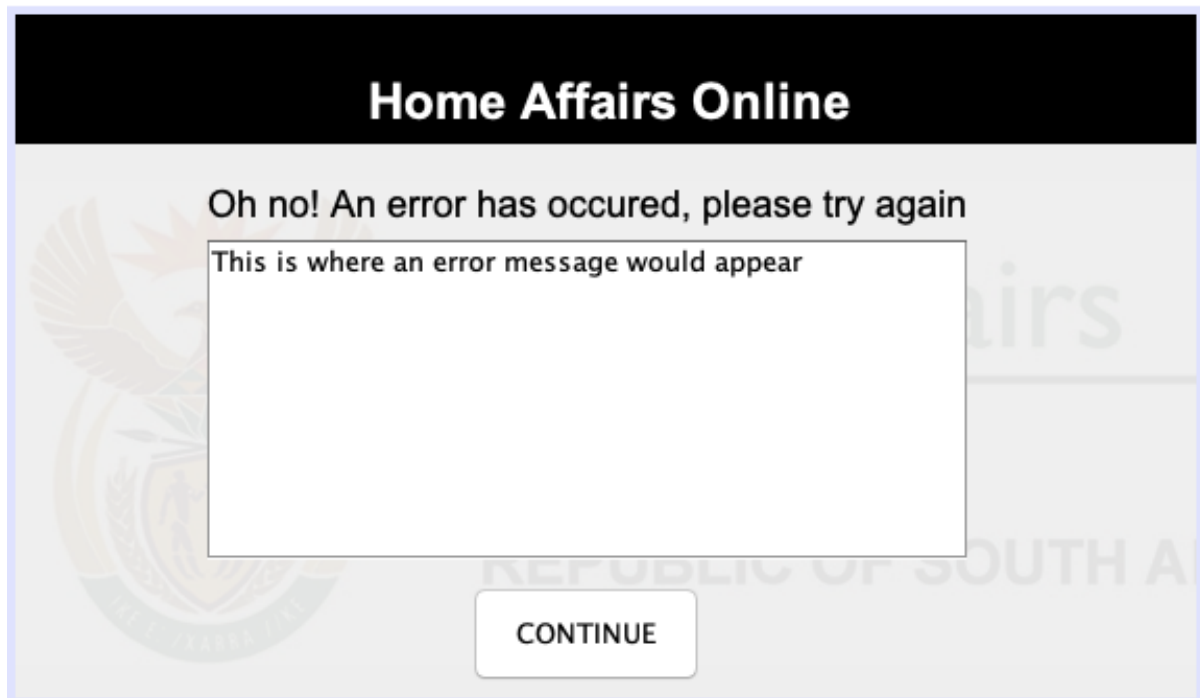
Security Group	All applicants, including the applicants who do not have log in details, can access this screen.
Error Checking	If no data is entered or data is only entered into one of the two fields, then the program will simply display the error message instead of crashing or bringing up a non-applicant friendly error message.

1.3 Action Performed Screen



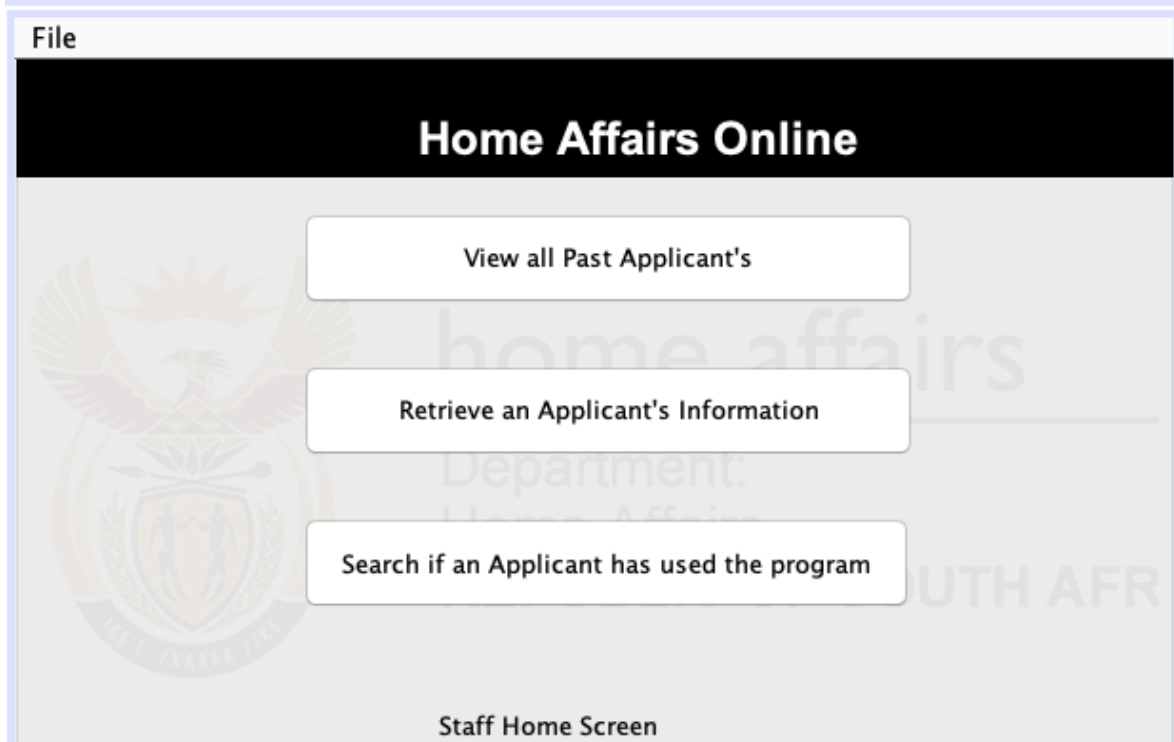
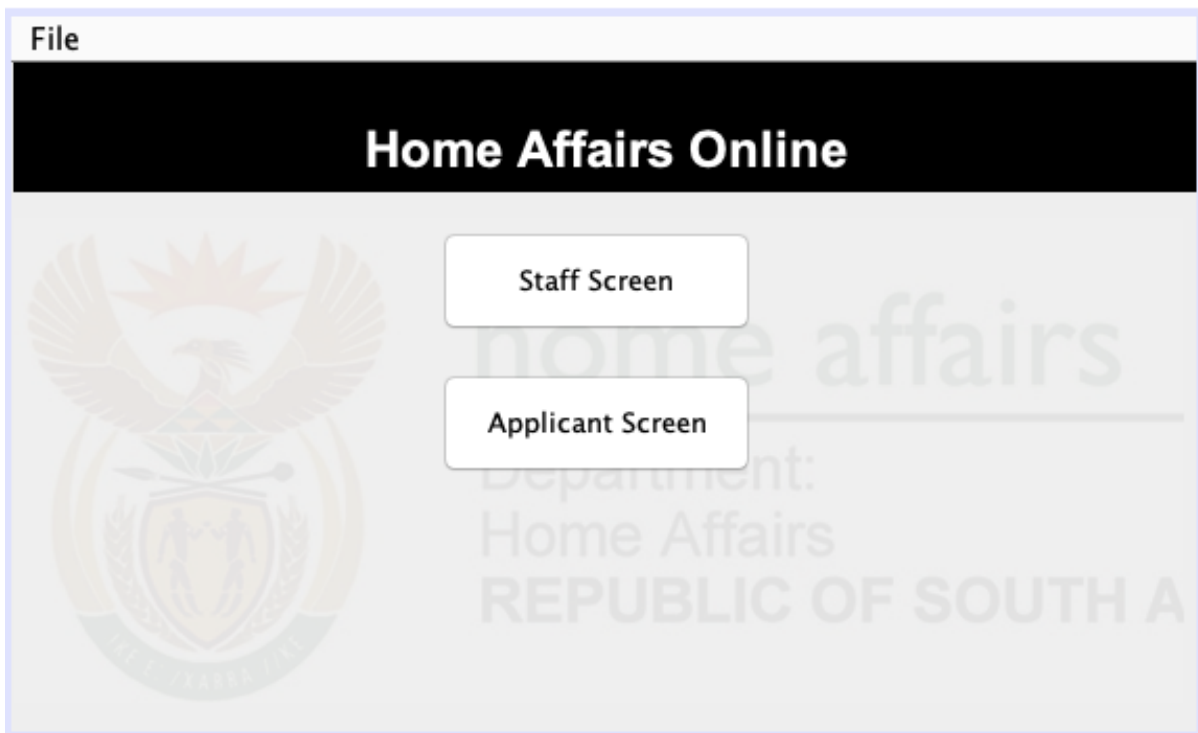
Input	Continue button – this may be clicked to take the applicant to the correct home screen.
Output	The above message and icon will be displayed to the applicant. This is an indication that the program has executed the action asked by the applicant correctly.
Processing	No processing is required in this screen as there is no data being manipulated, only a change of screens occur. Once the Continue button is clicked, the program will check whether the applicant is a staff member or just an applicant.
Data Storage	No data is being stored on this screen as this screen just signifies that the action has been performed.
Security Group	All applicants with a valid ID and password can access this screen.
Error Checking	No error checking will occur on this screen.

1.4 Error Screen



Input	Continue Button – this may be clicked to take the applicant to the home screen.
Output	The above message and icon will be displayed to the applicant. This is an indication that the program has encountered an error whilst attempting to perform its function. An example of this is when the incorrect data has entered or if it was unable to access the database.
Processing	No processing is required in this screen as there is no data manipulation, only a change of screen occurs.
Data Storage	No data is being stored on this screen as this screen simply signifies that an error has occurred.
Security Group	All applicants can access this screen if an error has occurred.
Error Checking	No error checking will occur on this screen.

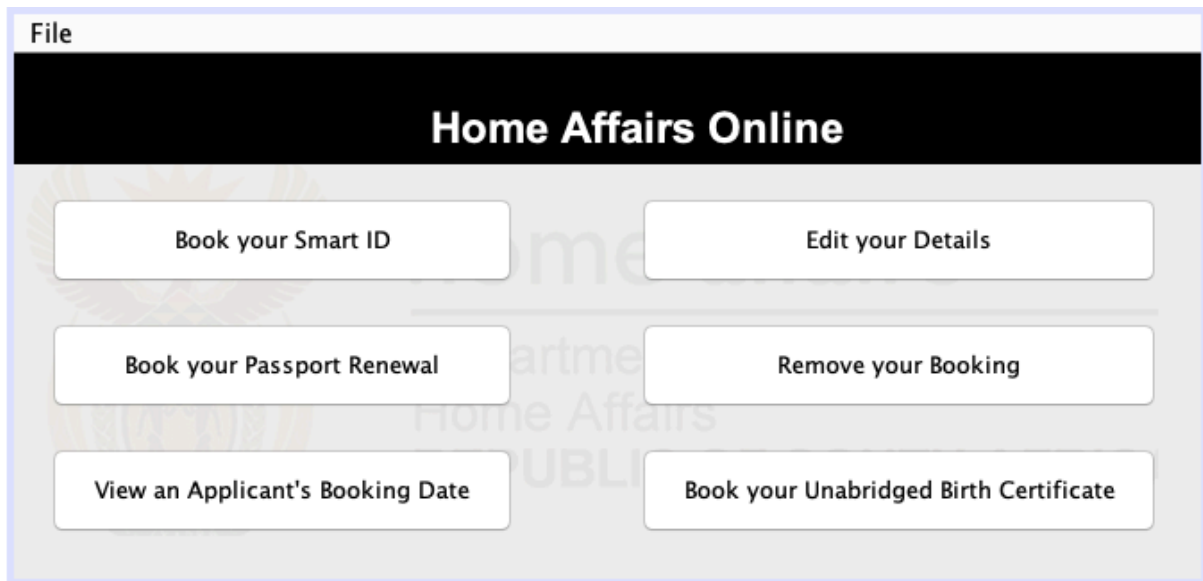
1.5 Staff Home Screen



Input	Staff Screen Button – This may be clicked to take the Staff member to the Staff Home Screen.
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	Applicant Button – This may be clicked to take the Staff member to the Applicant Home Screen View all Past Applicant's Button - This may be clicked to take the staff member to a list of all previous applicants which booked online through the website within each type of booking (Smart ID, Unabridged Birth Certificate, Passport Renewal). Retrieve an Applicant's Information Button - This may be clicked to view the information of an Applicant. Search if an Applicant has used the program Button – This may be clicked to view if a user has used the Home Affairs program before. File - This may be used to bring up a log out option.
Output	When the staff member selects a button, depending on the button a new screen will appear.
Processing	No processing is required in this screen as there is no data manipulation, only a change of screen occurs.
Data Storage	No data is being stored on this screen as this screen simply allows staff members to access a new screen.
Security Group	Only Staff members have access to this screen
Error Checking	No error checking will occur on this screen.

1.6 Applicant Home Screen



Input	Book a Smart ID Button – This may be clicked to take the applicant to the smart ID booking page so the applicant can book a date to go to the bank as well as enter the necessary information needed. Book a Passport Renewal Button - This may be clicked to take the applicant to the Passport Renewal booking page so the applicant can book a date to go to the bank as well as enter the necessary information needed. View Applicant's Booking Date Button - This may be clicked to take the applicant to a page which shows the date in which they have booked their appointment at the bank. Book a Unabridged Birth Certificate Button - This may be clicked to take the applicant to the birth certificate booking page so they can book a date to go to the bank as well as enter the necessary information needed.
--------------	--

	Edit Details Button - This may be clicked to take the applicant to a page where they can edit their personal information. Remove Booking Button - This may be clicked to remove an applicant's booking. File - This may be used to bring up a log out option.
Output	When the applicant selects a button, depending on the button a new screen will appear, such as the ID booking screen if the Book a Smart ID button is clicked.
Processing	No processing is required in this screen as there is no data manipulation, only a change of screen occurs.
Data Storage	No data is being stored on this screen as this screen simply allows applicants to access a new screen.
Security Group	Only applicants have access to this screen
Error Checking	No error checking will occur on this screen.

1.7 Smart ID Screen

File

Home Affairs Online

Enter your ID

Enter your full name

Enter your Surname

Are you a first time applicant? ☐ (Click box if yes)

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

Booking your Smart ID

File

Home Affairs Online

Photo of Certified copy of the Parent/ legal guardian ID

The photo will be inserted into this table.

Booking your Smart ID

File

Home Affairs Online

Photo of Certified copy of Birth Certificate

The photo will be inserted into this table.

BACK

Booking your Smart ID

CONTINUE

File

Home Affairs Online

Photo of proof of Residence

The photo will be inserted into this table.

BACK

Booking your Smart ID

CONTINUE

File

Home Affairs Online

Photo of Certified copy of passport

The photo will be inserted into this table.

BACK

Booking your Smart ID

CONTINUE

Input	Continue Button – this may be clicked to take the applicant to the next screen, the final continue button will change to the booking screen so the applicant can book a time to visit the bank. Back Button – this may be clicked to take the applicant to the previous screen. Check Box – This may be clicked to see if the applicant is a first time applicant or is returning, if the box is checked, the user doesn't have to put a photo of their parent/ legal gaurdians ID. Editor Pane – This is where the applicant can place their photo which will change the name of the photo to the next available number (such as image100.jpg) and store the name of the photo in a String and the actual photo in a folder with the respective name.
-------	--

	Text Field – The applicant may enter their ID, Full Name and surname into the respective fields. File - This may be used to bring up a log out option.
Output	When the final continue button is clicked, the applicant will be able to book a date in which they go to the bank.
Processing	In the background, the full name and surname will be checked to match the full name and surname which is contained with the ID, the information will then be stored.
Data Storage	The information provided by the applicant will be stored in the table tblSmartID as well as the photos being stored in the respective folders.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The name and surname will be checked to ensure it matches the ID number in the text file. • The Editor pane's will ensure that only photos in the form of .jpg are used. • The ID will be checked to be in the database.

1.8 Passport Renewal Screen

File

Home Affairs Online

Enter your ID

Enter your full name

Enter your Surname

Booking your Passport Renewal

File

Home Affairs Online

Photo of Certified copy of ID

The photo will be inserted into this table.

Booking your Passport Renewal

Input	Continue Button – this may be clicked to take the applicant to the next screen, after the final continue button the
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	applicant or staff member will be taken to a screen where they can choose the date and time of their booking. Back Button – this may be clicked to take the applicant to the previous screen. Editor Pane – This is where the applicant can place their photo which will change the name of the photo to the next available number (such as image100.jpg) and store the name of the photo in a String and the actual photo in a folder with the respective name. Text Field – The applicant may enter their ID, Full Name and surname into the respective fields. File - This may be used to bring up a log out option.
Output	When the final continue button is clicked, the applicant will be able to book a date in which they go to the bank as the screen will change.
Processing	In the background, the full name and surname will be checked to match the ID, the information will then be stored.
Data Storage	The information provided by the applicant will be stored in the table tblPassport and the photo will be stored in the appropriate folder and will be renamed.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The name and surname will be checked to ensure it matches the ID number. • The Editor pane's will ensure that only photos in the form of .jpg are used. • The ID will be checked to be in the database.

1.9 View a Booking Screen

File

Home Affairs Online



Enter your ID

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

BACK

Viewing your booking

CONTINUE

File

Home Affairs Online

ID	Name	Surname	Date	Time	Reason
030223609...	Byron	Kraus	Wed 01 April	11:00AM	PassPort Re...

BACK

View a booking

FINISHED

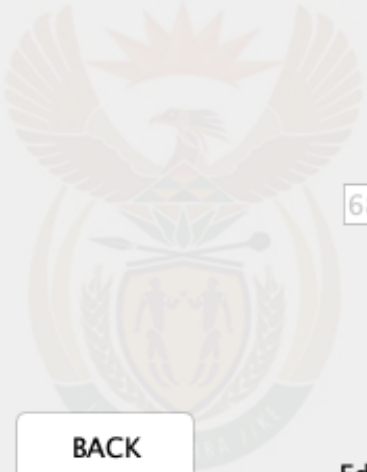
Input	Continue Button – this may be clicked to take the applicant to the next screen.
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	Finished Button – this may be clicked to take the applicant back to the applicant home screen or staff home screen. Back Button – this may be clicked to take the applicant to the previous screen. Text Field – The applicant may enter their ID into the respective field. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the applicant will be able to view the date that they booked as well as all their personal details, they can then click finished to return back to the applicant home screen or the staff home screen.
Processing	No storing will occur. The information inputted will be used to find the applicants booking.
Data Storage	No information will be stored, the information inputted will simply be used to find the corresponding information for the applicant or staff member to view.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The ID will be checked to be in the database.

1.10 Edit an applicant's details Screen

File

Home Affairs Online



Enter your ID

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

Edit the details of your booking

File

Home Affairs Online

Enter your ID ☐

Enter your full name ☐

Enter your middle name ☐

Enter your Surname ☐

Enter your cell number ☐

Enter your email ☐

Correct the incorrect details of your/ the users name, surname and/or ID and click the corresponding round button to change it.

Edit a user's details

Input	Continue Button – this may be clicked to take the applicant to the next screen.
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	Back Button – this may be clicked to take the applicant to the previous screen. Text Field – The applicant may enter their ID into the respective field for the first screen and edit their details in the second screen. Radio Button – this may be clicked to allocate that changes have been made to the corresponding information on the left of it. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the applicant will be able to view their personal information, they can then edit anything if it is incorrect, they may not change their information to whatever they want as when they go to the bank to book for something if their personal information is incorrect and they don't have a certified copy exclaiming that they got those details legally changed they will have their information changed back.
Processing	The information inputted will be used to find the applicants booking, they can then change the information and when they do the information will be updated in tblApplicants.
Data Storage	The information being updated will be stored in tblApplicants.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The ID will be checked to be in the database

1.11 Remove a booking Screen

File

Home Affairs Online

Enter your ID

Department: Home Affairs
REPUBLIC OF SOUTH AFRICA

BACK

CONTINUE

Remove a booking

Input	Continue Button – this may be clicked to take the applicant to the next screen. Back Button – this may be clicked to take the applicant to the previous screen. Text Field – The applicant may enter their ID into the respective field. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the applicant will be able to view the date they have booked for their appointment, they can then delete the booking if they cannot make the appointment, or they have made an error.

Processing	No storing will occur. The information inputted will be used to find the applicants booking so it can be deleted if they confirm.
Data Storage	No information will be stored, the information will simply be used for the next page.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The ID will be checked to contain a booking.

1.12 Confirm Remove a Booking Screen

File

Home Affairs Online

ID	Name	Surname	Date	Time	Reason
030223609...	Byron	Kraus	Wed 01 April	11:00AM	Smart ID

BACK

CONFIRM

Remove a booking

Input	Confirm Button – this may be clicked to remove the booking and take the applicant back to the applicant main menu or staff main menu. Back Button – this may be clicked to take the applicant to the previous screen. Table – Any specific row may be chosen, this is the date and time in which they are booking to go to the bank. File - This may be used to bring up a log out option.
Output	When the confirm button is clicked, the screen will change back to the applicant home screen or the staff home screen.
Processing	In the background the booking will be made available and the applicants booking will be removed as well as all the information they entered for that booking.

Data Storage	No storage will take place as data is being removed.
Security Group	All applicants can access these screen.
Error Checking	No error checking will take place.

1.13 Unabridged Birth Certificate Screen

The screenshot shows a web interface for booking an unabridged birth certificate. At the top, there is a 'File' tab and a black header with the text 'Home Affairs Online'. Below the header, there are four text input fields with labels: 'Enter your ID' (containing '6801023487265'), 'Enter your full name' (containing 'John'), 'Enter your Surname' (containing 'Williamson'), and 'Enter a parent/Guardian's ID' (containing '5405053347265'). At the bottom left is a 'BACK' button, and at the bottom right is a 'CONTINUE' button. The background features a faint watermark of the South African coat of arms and the text 'Home Affairs Department' and 'REPUBLIC OF SOUTH AFRICA'. At the very bottom, the text 'Booking your Unabridged Birth Certificate' is displayed.

Input	Continue Button – this may be clicked to take the applicant to the next screen where they can book the date in which they going to go to the bank. Back Button – this may be clicked to take the applicant to the previous screen. Text Field – The applicant may enter their ID, full name, surname and the ID of a parent or guardian into the respective fields. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the applicant will be able to make the date in which they booked to go to the bank.
Processing	The information inputted will be stored and then the applicant will be able to book a date to go to the bank.

Data Storage	The information will be stored into tblUnabridgedBirthCertificate.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The ID and the parents ID will be checked to only contain numbers. • The name and surname will be checked to ensure it matches the ID number. • The ID will be checked to be on the database.

1.14 Select a Booking Screen

Date	Time
Wed 01 April	10:00AM
Thu 02 April	10:00AM
Thu 02 April	11:00AM
Fri 03 April	09:00AM
Fri 03 April	9:00AM
Fri 03 April	10:00AM
Fri 03 April	11:00AM
Fri 03 April	01:00PM

BACK CONTINUE

Selecting your booking

Input	Continue Button – this may be clicked to confirm which booking they have chosen and return to the applicant home screen or staff home screen. Back Button – this may be clicked to take the applicant to the previous screen. Table – Any specific row may be chosen, this is the date and time in which they are booking to go to the bank. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the applicant will be returned back to the applicant home screen or staff home screen.

Processing	In the background the date and time chosen will be saved into the respective table with the applicant which booked it and that booking will no longer be available.
Data Storage	The information will be stored into tblBooking.
Security Group	All applicants can access these screen.
Error Checking	The following will be checked: • The date and time will be checked so that it isn't booked. - If that booking isn't taken, it will be booked for the applicant - If that booking is taken, an error message will appear and the applicant will be able to book another date.

1.15 View All Past Applicant's Screen

File

Home Affairs Online

Reason for Booking
Smart ID
UnaBridged Birth Certificate
Passport

BACK

select the type of booking

CONTINUE

File

Home Affairs Online

ID	Name	Surname	Date	Time
0302236096099	Byron	Kraus	Wed 01 April	11:00AM
0204050404040	James	Matthews	Wed 02 April	11:00AM

BACK

All Past users

FINISHED

Input	Continue Button – this may be clicked to take the applicant to the next screen. Finished Button – this may be clicked to take the applicant back to the staff main menu. Back Button – this may be clicked to take the applicant to the previous screen. Table – on the first screen the staff member may chose the type of booking he wants to view the past applicants on. File - This may be used to bring up a log out option.
Output	When the confirm button is clicked, the staff member will return back to the staff home screen.
Processing	In the background the correct table will be chosen to allow the staff member to see all the past applicants.
Data Storage	No storage will take place.
Security Group	Only staff members can access these screen.
Error Checking	No error checking will take place.

1.16 Retrieve an Applicant's Information Screen

File

Home Affairs Online

Reason for Booking

Smart ID

UnaBridged Birth Certificate

Passport

BACK

Retrieve information of booking

CONTINUE

File

Home Affairs Online

Enter your ID

6801023487265

Department:
Home Affairs
REPUBLIC OF SOUTH A

BACK

Retrieve information of booking

CONTINUE

File

Home Affairs Online

ID	Name	Surname	Copy of ID
0302236096099	Byron	Kraus	image5166.jpg

BACK

Retrieve Passport information of booking

FINISH

File

Home Affairs Online

ID	Name	Surname	Birth Certificate	Copy of Paren...	Proof of Resid...
030223609...	Byron	Kraus	image345.jpg	image345.jpg	image345.jpg

BACK

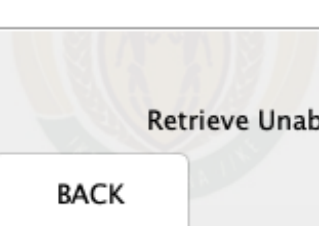
Retrieve Smart ID information of booking

FINISH

File

Home Affairs Online

ID	Name	Surname	Parents ID number
0302236096099	Byron	Kraus	6602014074077



Home Affairs
REPUBLIC OF SOUTH AFRICA

Retrieve Unabridged Birth Certificate information of booking

BACK

FINISH

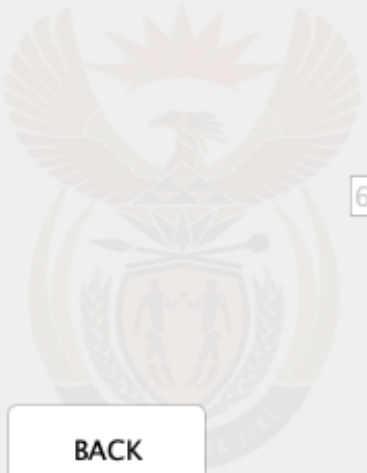
Input	Confirm Button – this may be clicked to take the applicant to the next screen. Finish Button – this may be clicked to take the staff member back to the staff main member. Back Button – this may be clicked to take the applicant to the previous screen. Table – on the first screen the staff member may chose the type of booking they want to view. Text Field – The staff member may enter the applicants ID into the respective field. File - This may be used to bring up a log out option.
Output	When the confirm button is clicked, the staff member will return back to the staff home screen.

Processing	In the background the correct table will be chosen to allow the staff member to see the applicants information.
Data Storage	No storage will take place.
Security Group	Only staff members can access these screens.
Error Checking	• The ID will be checked to only contain numbers. • The ID will be checked to be on the database.

1.17 Search User Screen

File

Home Affairs Online



Enter your ID

6801023487265

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

BACK

CONTINUE

Search for Applicant booking

File

Home Affairs Online

ID	Name	Surname	Date	Time	Reason
030223609...	Byron	Kraus	Wed 01 April	11:00AM	PassPort Re...

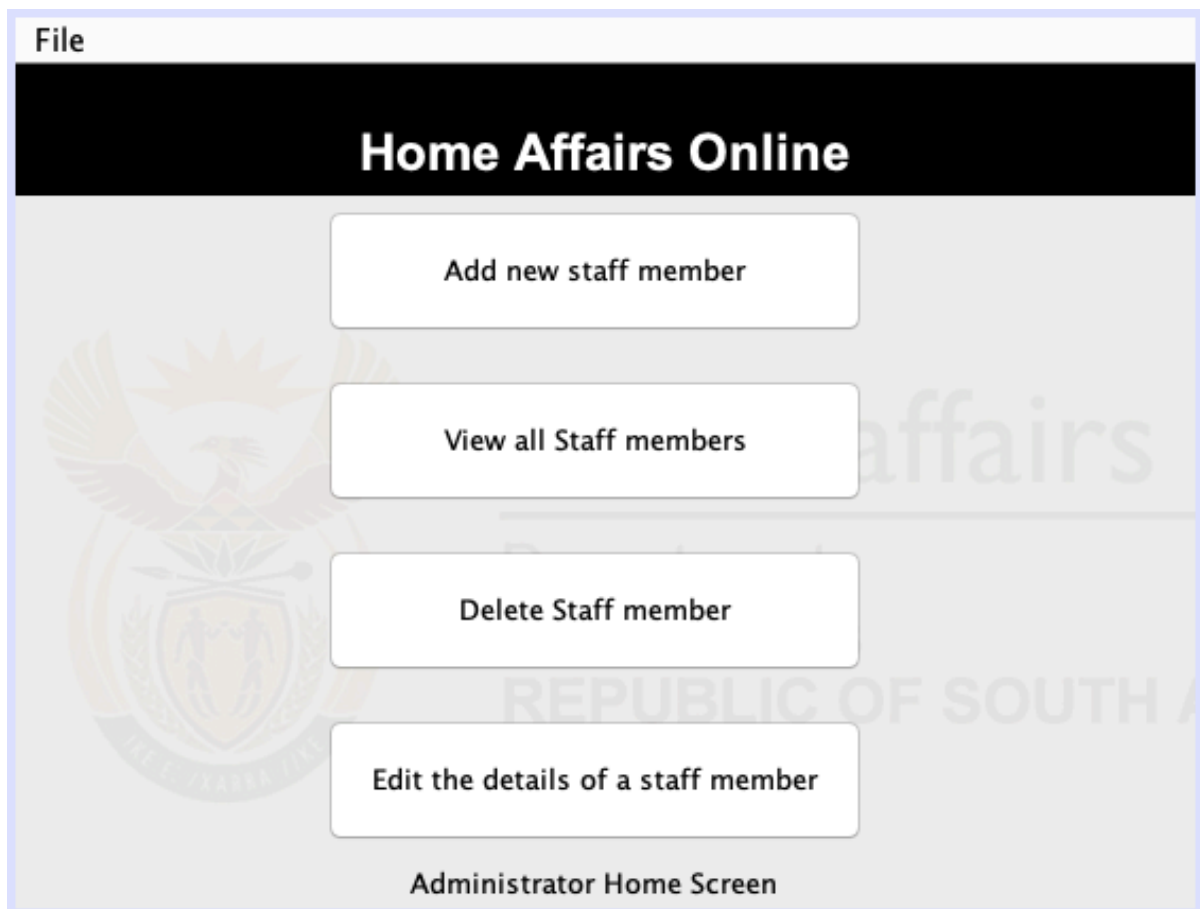
BACK

Search Applicant

FINISHED

Input	Continue Button – this may be clicked to take the Staff Member to the next screen. Finished Button – this may be clicked to take the staff member back to the staff main member. Back Button – this may be clicked to take the Staff member to the previous screen. Text Field – The staff member may enter the applicants ID into the respective field. File - This may be used to bring up a log out option.
Output	When the confirm button is clicked, the staff member will return back to the staff home screen.
Processing	In the background the correct table will be chosen to allow the staff member to see the applicants information.
Data Storage	No storage will take place.
Security Group	Only staff members can access these screens.
Error Checking	• The ID will be checked to only contain numbers. • The ID will be checked to be on the database.

1.18 Administrator Home Screen



Input	Add new staff member Button – This may be clicked to take the administrator to a screen where they can add the details of the new staff member. Edit details of Staff member Button – This may be clicked to take the administrator to a screen where they can edit the details of a staff member. Delete Staff member Button – This may be clicked to take the administrator to a page which allows them to delete a staff member that no longer works for home affairs. View all Staff members Button – This may be clicked to take the administrator to the page where they can view all the staff members currently employed.
--------------	---

	File - This may be used to bring up a log out option.
Output	When the administrator selects a button, depending on the button a new screen will appear.
Processing	No processing is required in this screen as there is no data manipulation, only a change of screen occurs.
Data Storage	No data is being stored on this screen as this screen simply allows applicants to access a new screen.
Security Group	Only administrators have access to this screen
Error Checking	No error checking will occur on this screen.

1.19 Add a new Staff Member screen

File

Home Affairs Online

Enter your ID

6801023487265

Enter your role

0

(1 – Staff member)
(2 – Administrator)

Enter your full name

John

Enter your middle name

Fred

Enter your Surname

Williamson

Enter a password

.....

BACK

Add a new Staff member

CONTINUE

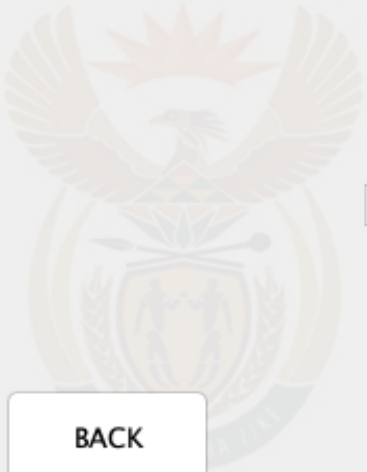
Input	Text Field – The administrator may enter the Staff members ID, full name, surname and password into the respective field. Continue Button – this may be clicked to take the administrator to the next screen. Back Button – this may be clicked to take the administrator to the previous screen. File - This may be used to bring up a log out option.
Output	When the administrator selects a button, depending on the button a new screen will appear.
Processing	No processing is required in this screen as there is no data manipulation, only a change of screen occurs.
Data Storage	No data is being stored on this screen as this screen simply allows applicants to access a new screen.
Security Group	Only administrators have access to this screen

Error Checking	No error checking will occur on this screen.
-----------------------	--

1.20 Delete a Staff member screen

File

Home Affairs Online



Enter your ID

Department:
Home Affairs
REPUBLIC OF SOUTH A

BACK

CONTINUE

Remove a Staff member

File

Home Affairs Online

ID	Name	Surname
0302236096099	Byron	Kraus

BACK

CONFIRM

Remove a Staff member

Input

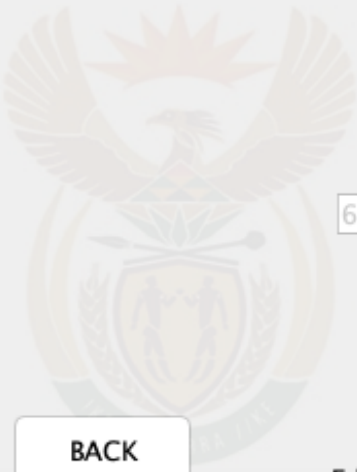
Continue Button – this may be clicked to take the administrator to the next screen.

	Back Button – this may be clicked to take the administrator to the previous screen. Text Field – The applicant may enter their ID into the respective field. Table – This field stores the name, surname and ID of the staff member. File - This may be used to bring up a log out option.
Output	When the continue Button is pressed the Administrator can view the staff member to make sure it is the correct staff member before pressing confirm and removing the Staff Member from the database.
Processing	No storing will occur. The information inputted will be used to find the Staff member so they can be deleted if they confirm.
Data Storage	No information will be stored, the information will simply be used for the next page.
Security Group	Only administrators can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The ID will be checked to be a staff member.

1.21 Edit a Staff member screen

File

Home Affairs Online



Enter your ID

Department:
Home Affairs
REPUBLIC OF SOUTH AFRICA

BACK

Edit the details of a Staff Member

CONTINUE

File

Home Affairs Online

Enter your ID ☐

Enter your full name ☐

Enter your middle name ☐

Enter your Surname ☐

Correct the incorrect details of your/ the users name, surname and/or ID and click the corresponding round button to change it.

BACK

Edit the details of a Staff Member

CONTINUE

Input	Continue Button – this may be clicked to take the administrator to the next screen.
--------------	--

	Back Button – this may be clicked to take the administrator to the previous screen. Text Field – The administrator may enter the Staff members ID into the respective field for the first screen and edit their details in the second screen. Radio Button – this may be clicked to allocate that changes have been made to the corresponding information on the left of it. File - This may be used to bring up a log out option.
Output	When the continue button is clicked, the administrator will be able to view their Staff members personal information, they can then edit anything if it is incorrect, they may not change their information to whatever they want as when they go to the bank to book for something if their personal information is incorrect and they don't have a certified copy exclaiming that they got those details legally changed they will have their information changed back.
Processing	The information inputted will be used to find the Staff Members booking, they can then change the information and when they do the information will be updated in tblStaff.
Data Storage	The information being updated will be stored in tblStaff.
Security Group	Only Administrators can access these screen.
Error Checking	The following will be checked: • The ID will be checked to only contain numbers. • The ID will be checked to be in the database

1.22 View all Staff Members Screen

File

Home Affairs Online

ID	Name	Surname
0302236096099	Byron	Kraus
0204050404040	James	Matthews

BACK

All active Staff Members

FINISHED

Input	Finished Button – this may be clicked to take the administrator back to the administrator main menu. Back Button – this may be clicked to take the applicant to the previous screen. Table – The administrator may view the Staff currently working at Home Affairs. File - This may be used to bring up a log out option.
Output	When the confirm button is clicked, the administrator will return back to the administrator home screen.
Processing	No processing will take place, the staff members details will simply be fetched.
Data Storage	No storage will take place.
Security Group	Only administrators can access these screen.
Error Checking	No error checking will take place.

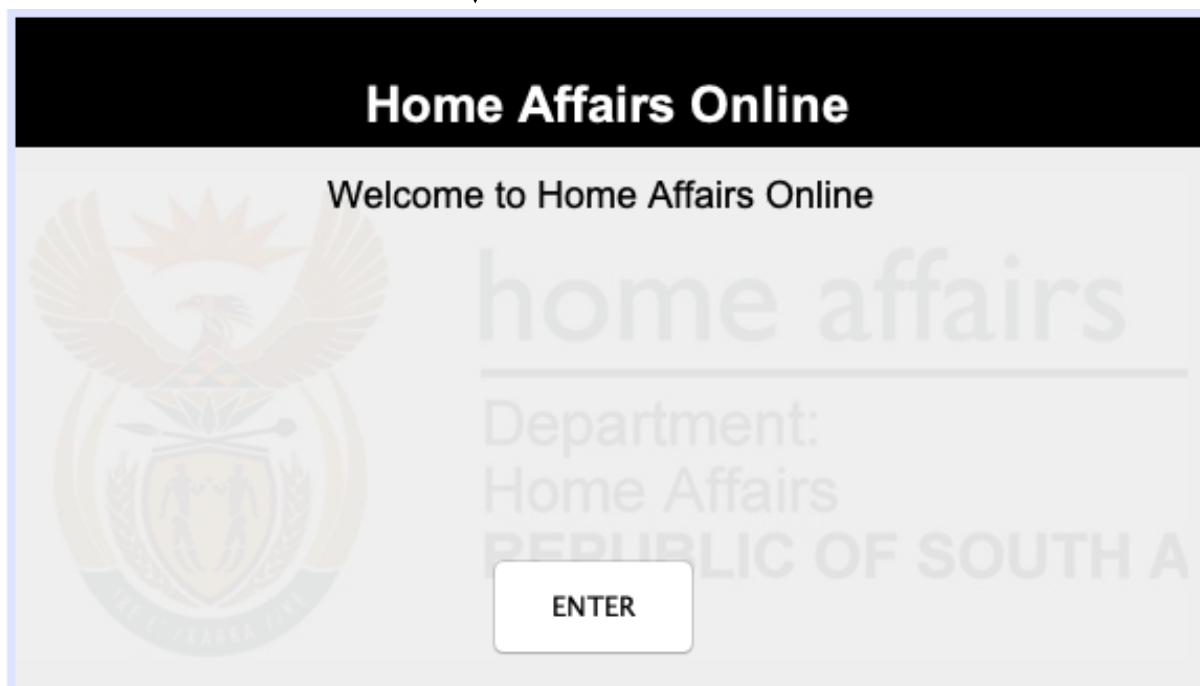
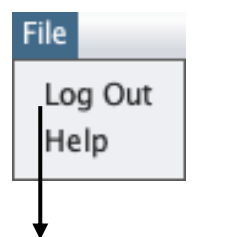
2 Sequencing

2.1 Introduction

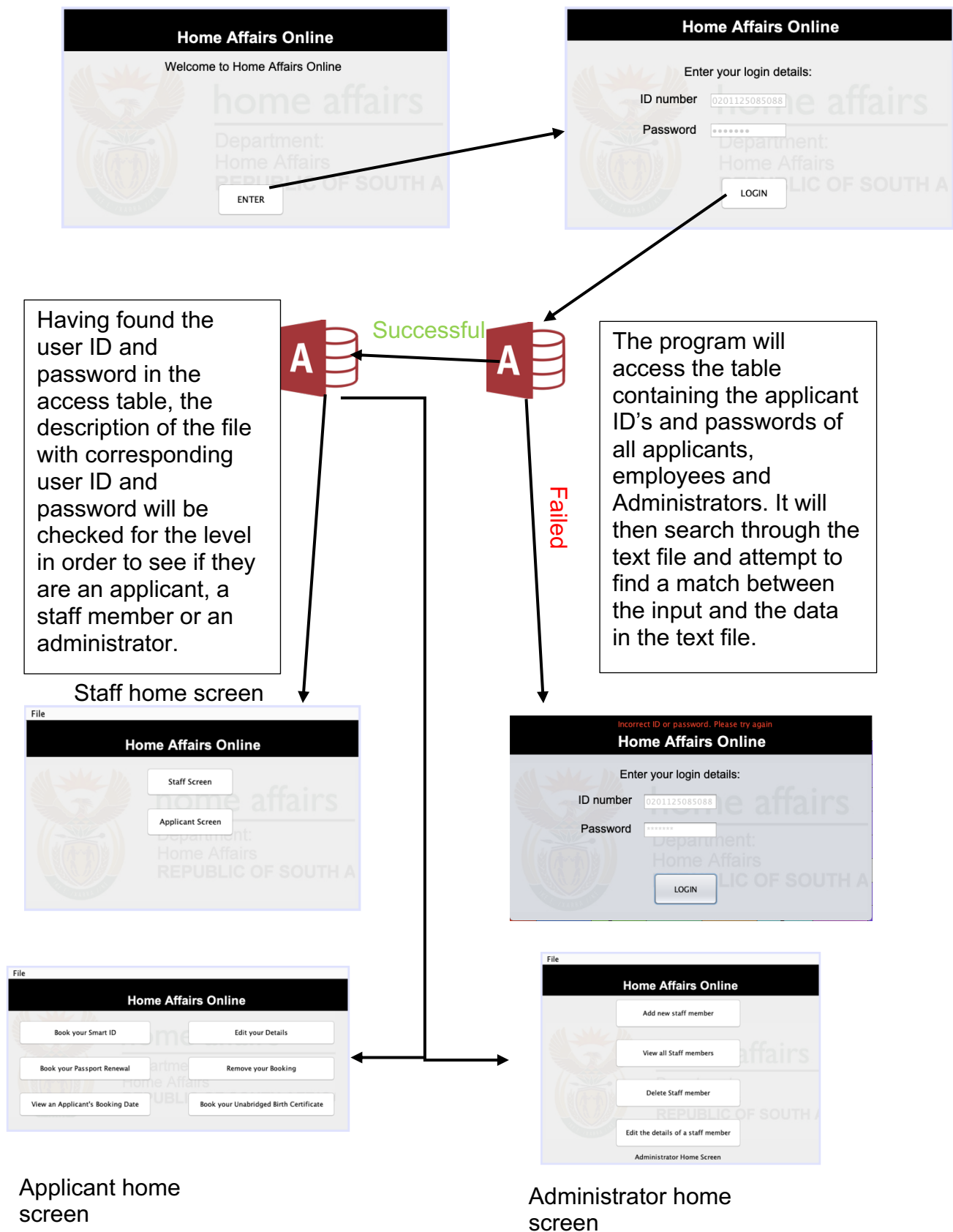
- If a problem arises due to data pointing towards non-existence data being entered by the applicant, the error screen will appear.
- If the file button is clicked, the menu list will appear. Certain screens will not have the file button as they do not need it such as the action performed screen and the welcome screen.

2.2 File Menu List

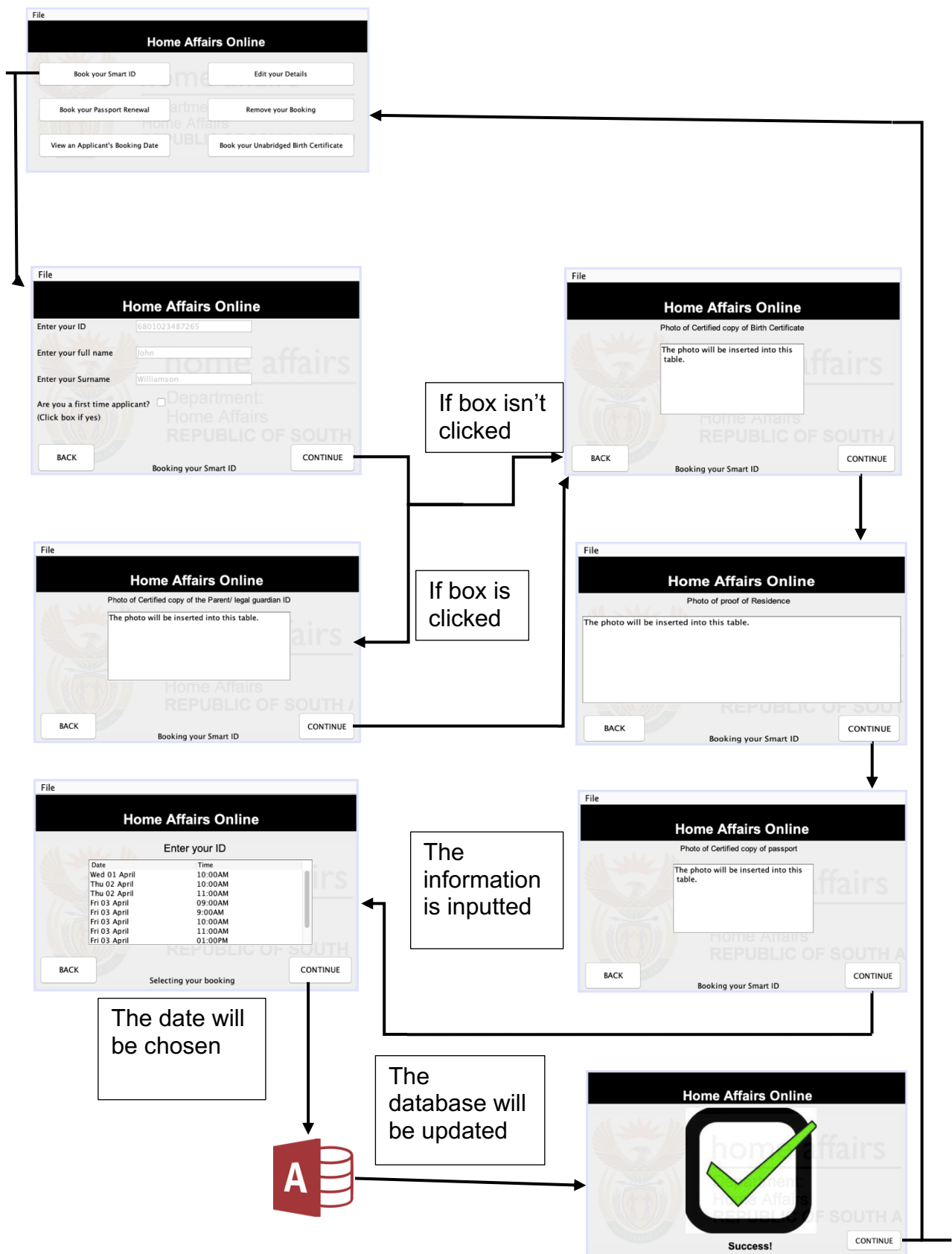
Generic version used for all screens once logged into their account.



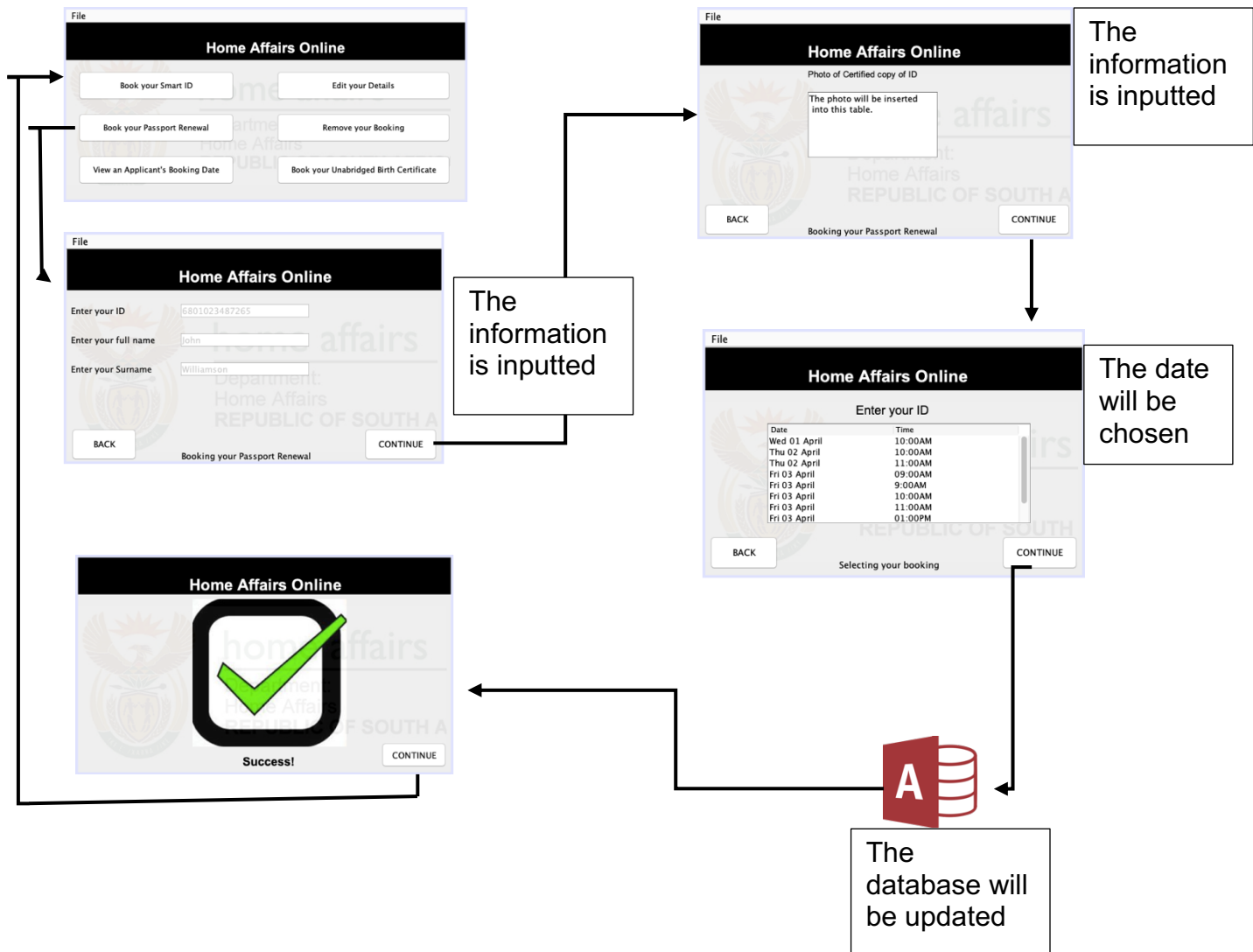
2.3 Login Screen



2.4 Smart ID screen

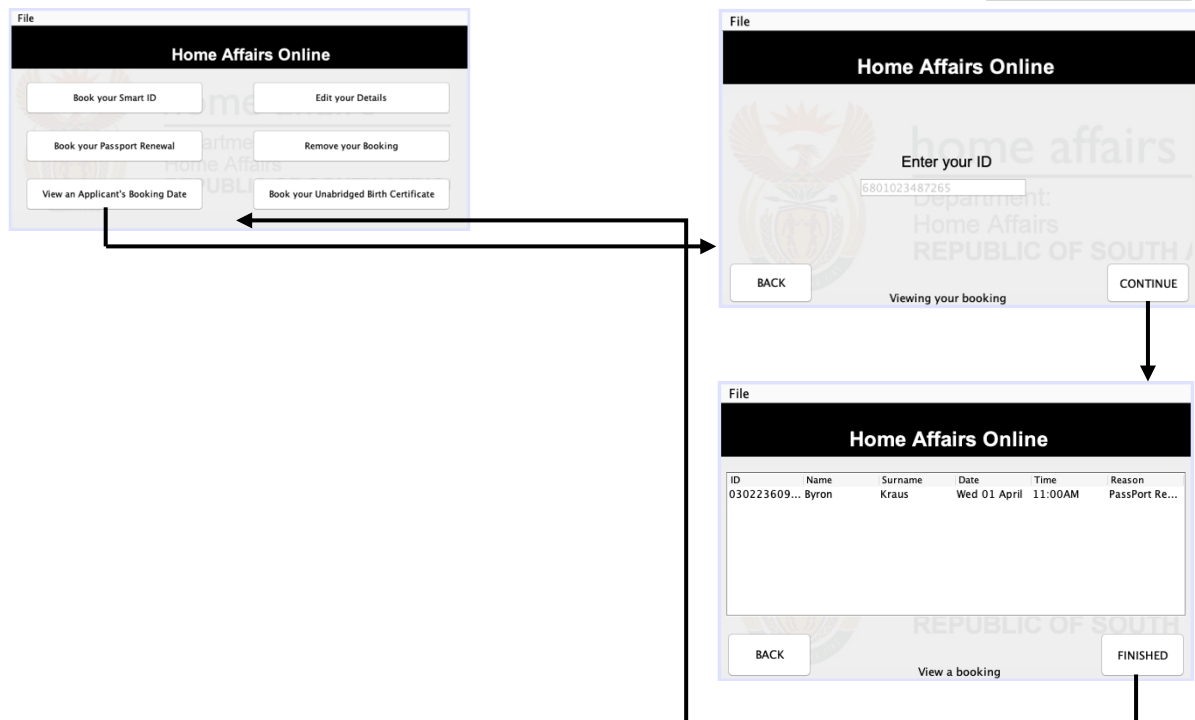


2.5 Passport Renewal Screen



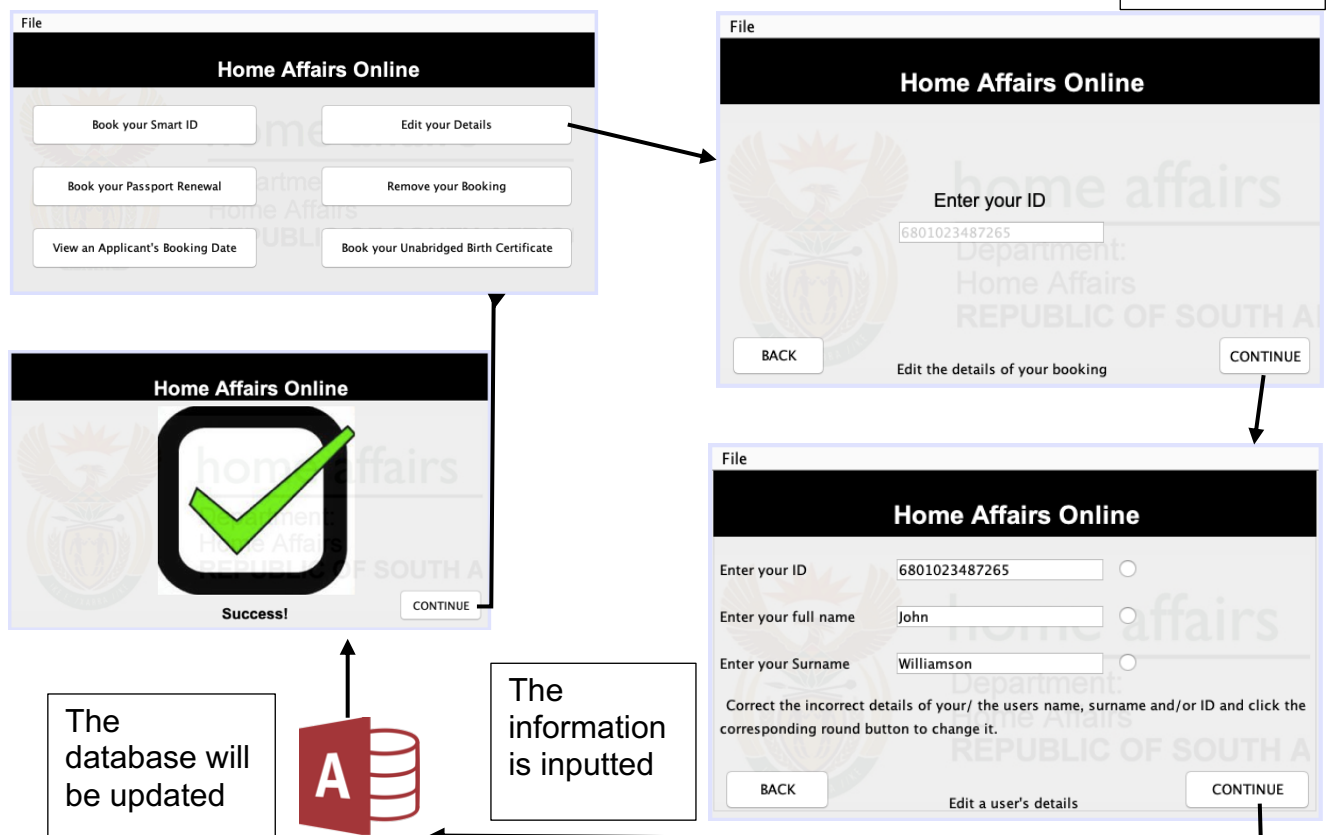
2.6 View an Applicant's Booking Screen

The information is inputted

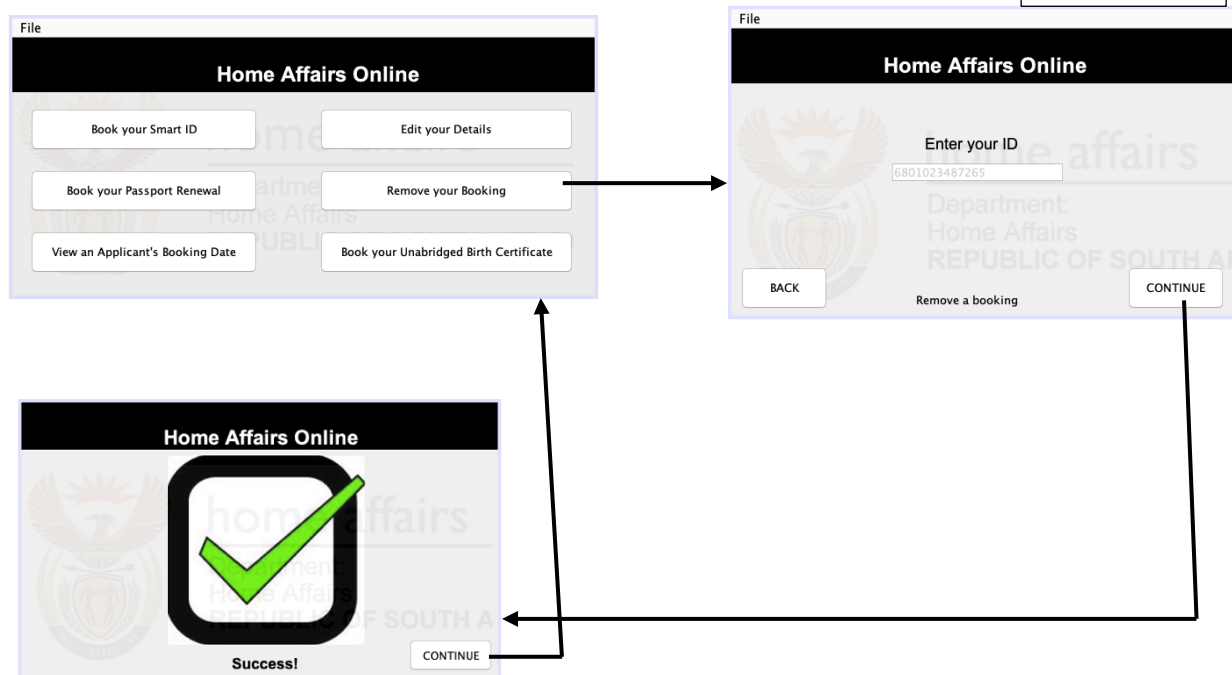


2.7 Edit your Details Screen

The information is inputted

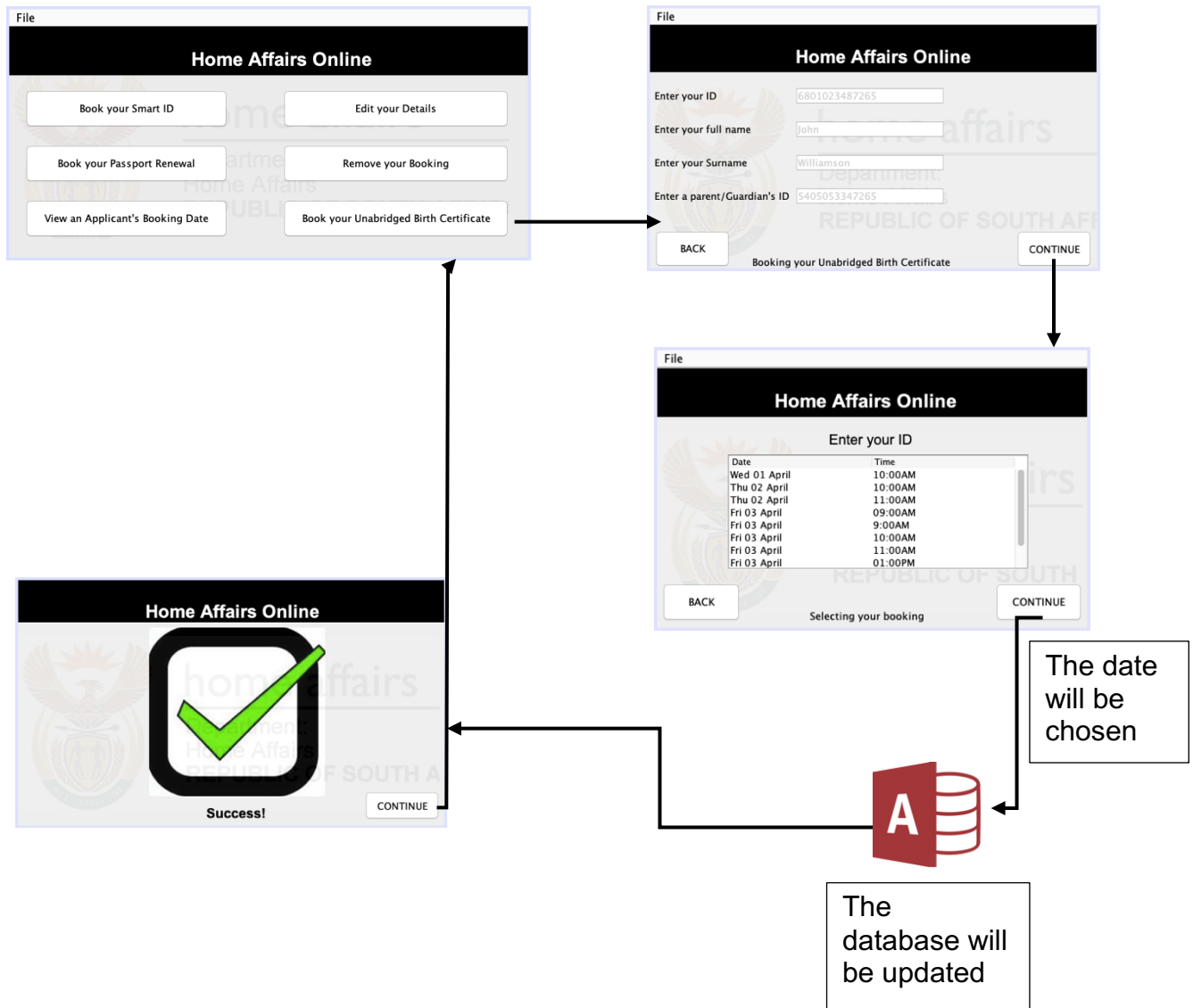


2.8 Remove your Booking Screen

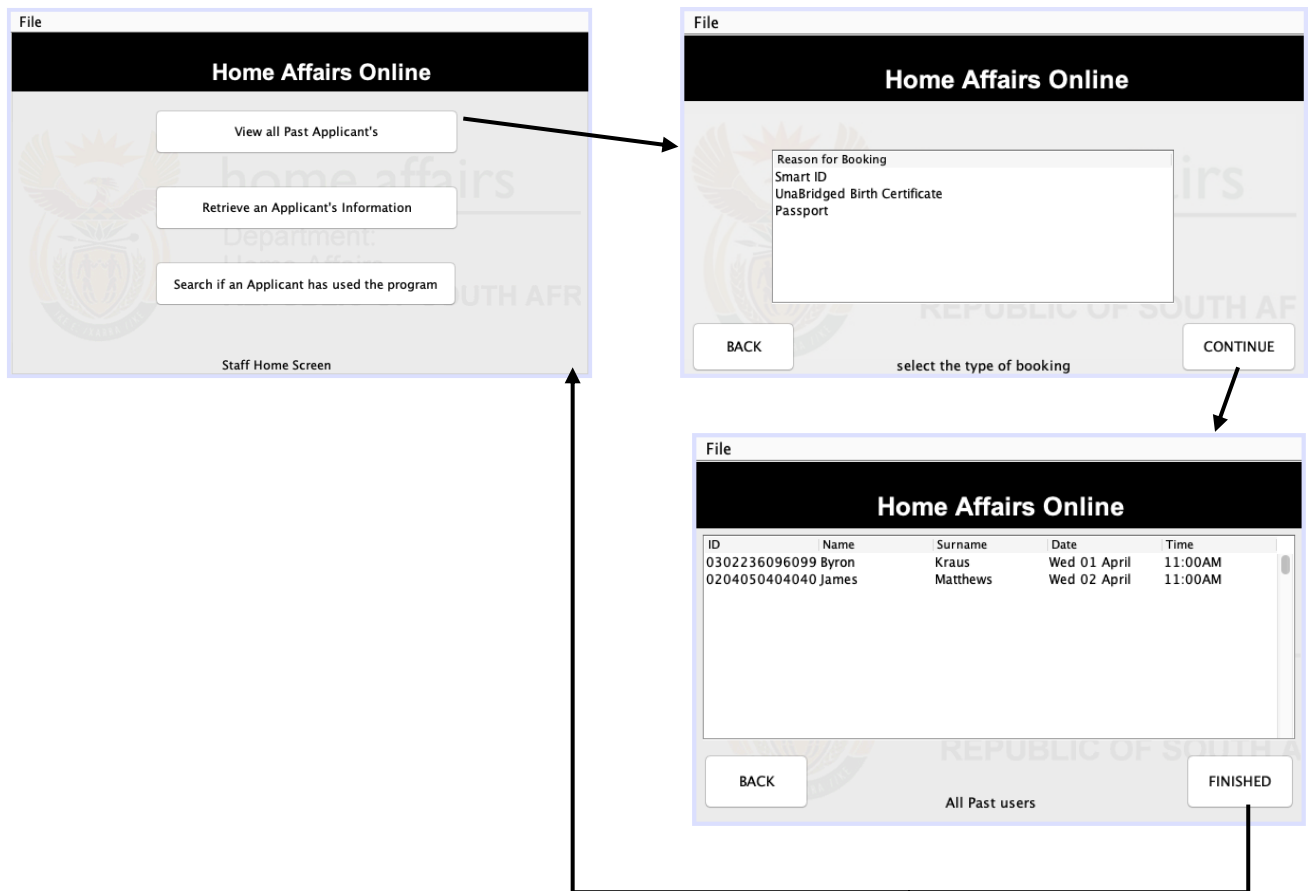


2.9 Book your Unabridged Birth Certificate Screen

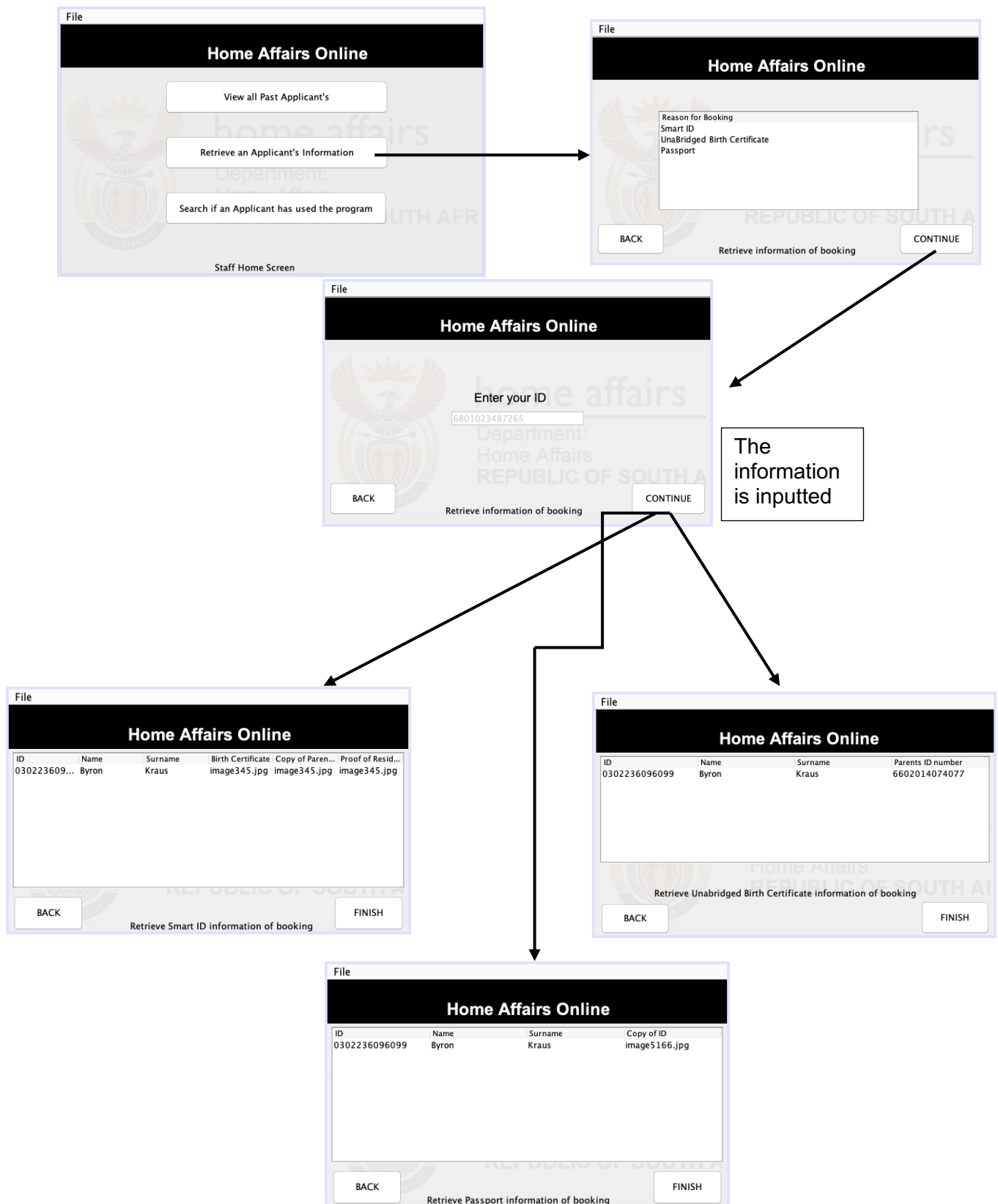
The information is inputted



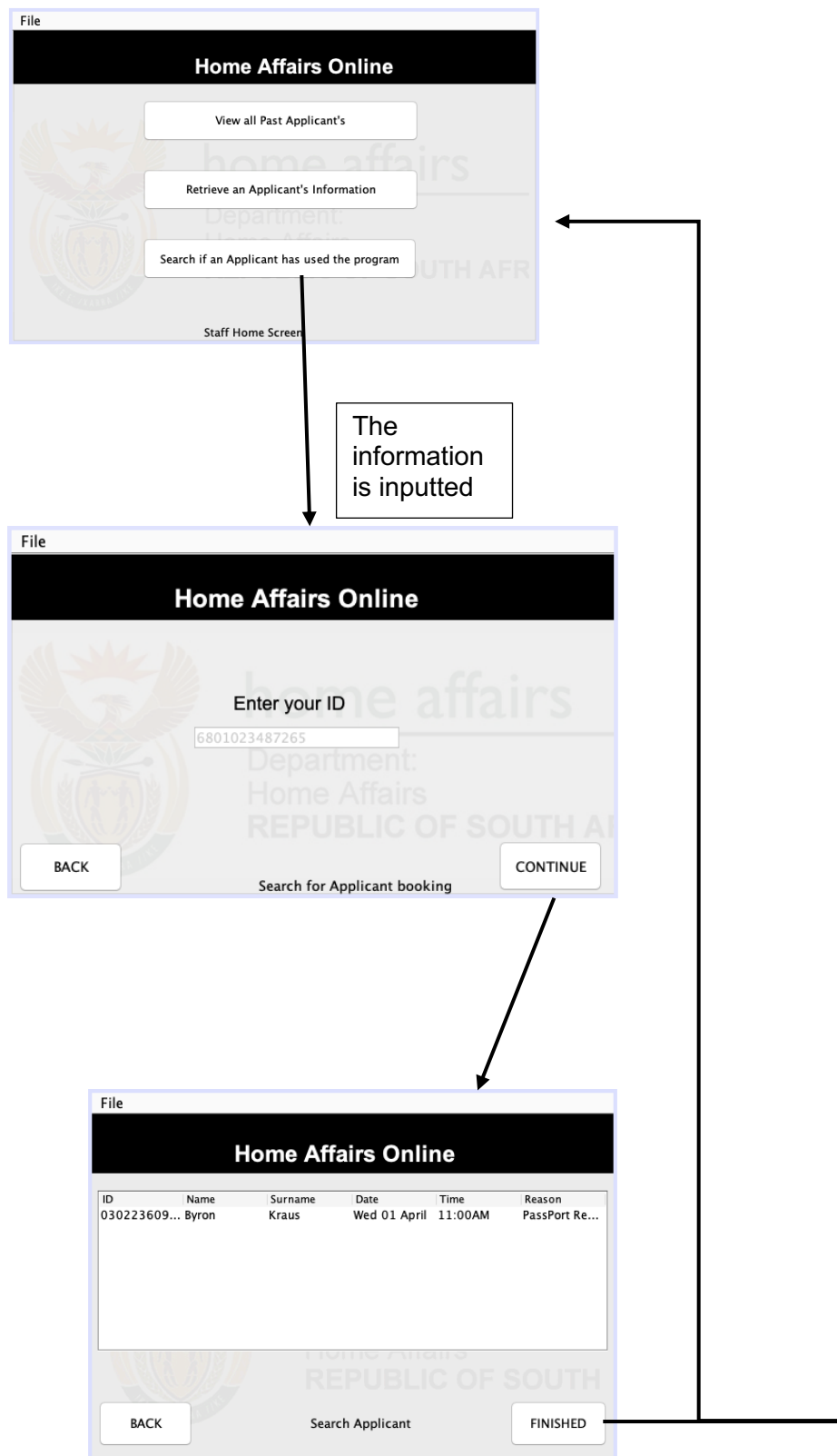
2.10 View all Past Applicant's Screen



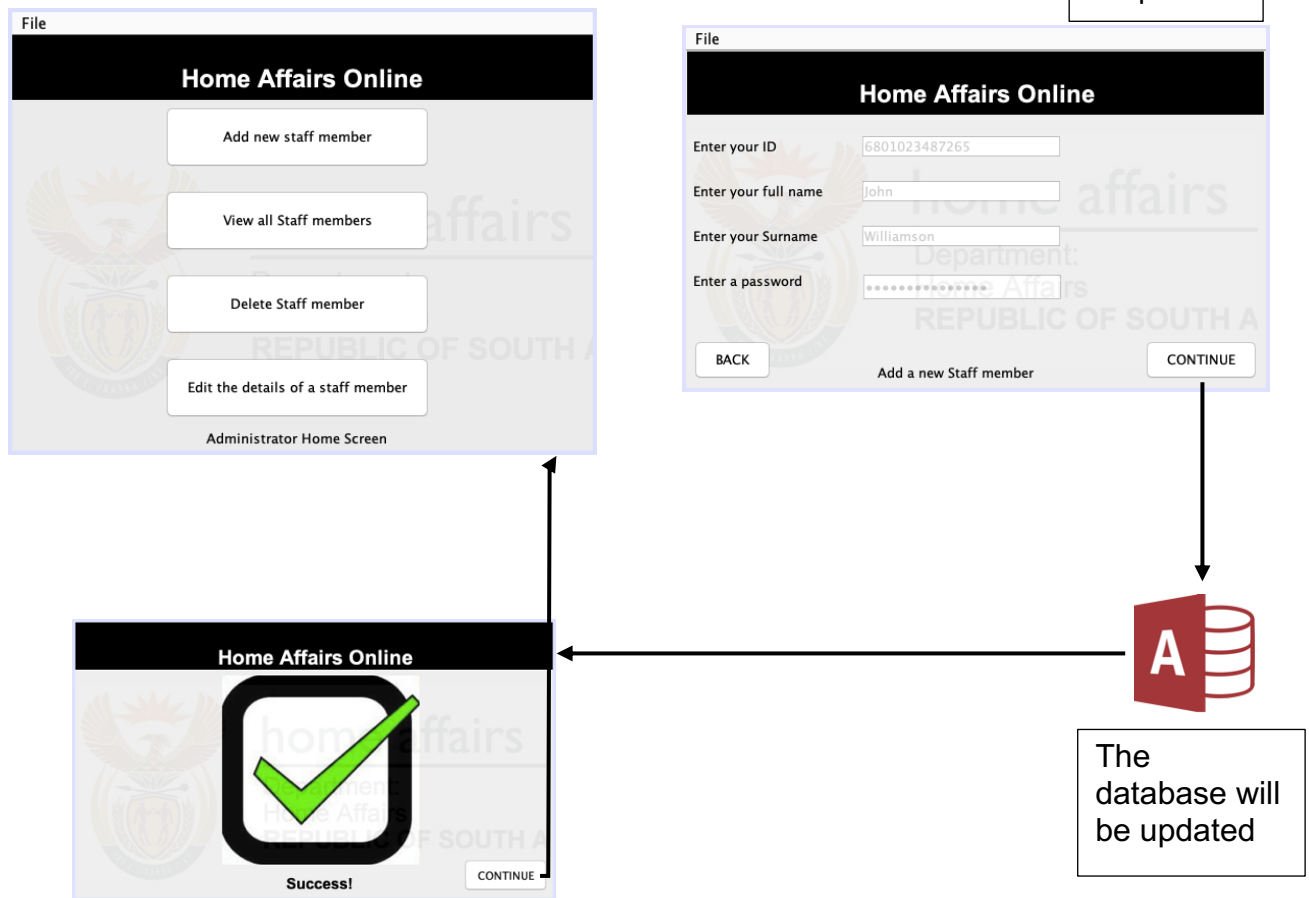
2.11 Retrieve an Applicant's Information Screen



2.12 Search if an Applicant Used the Program Screen



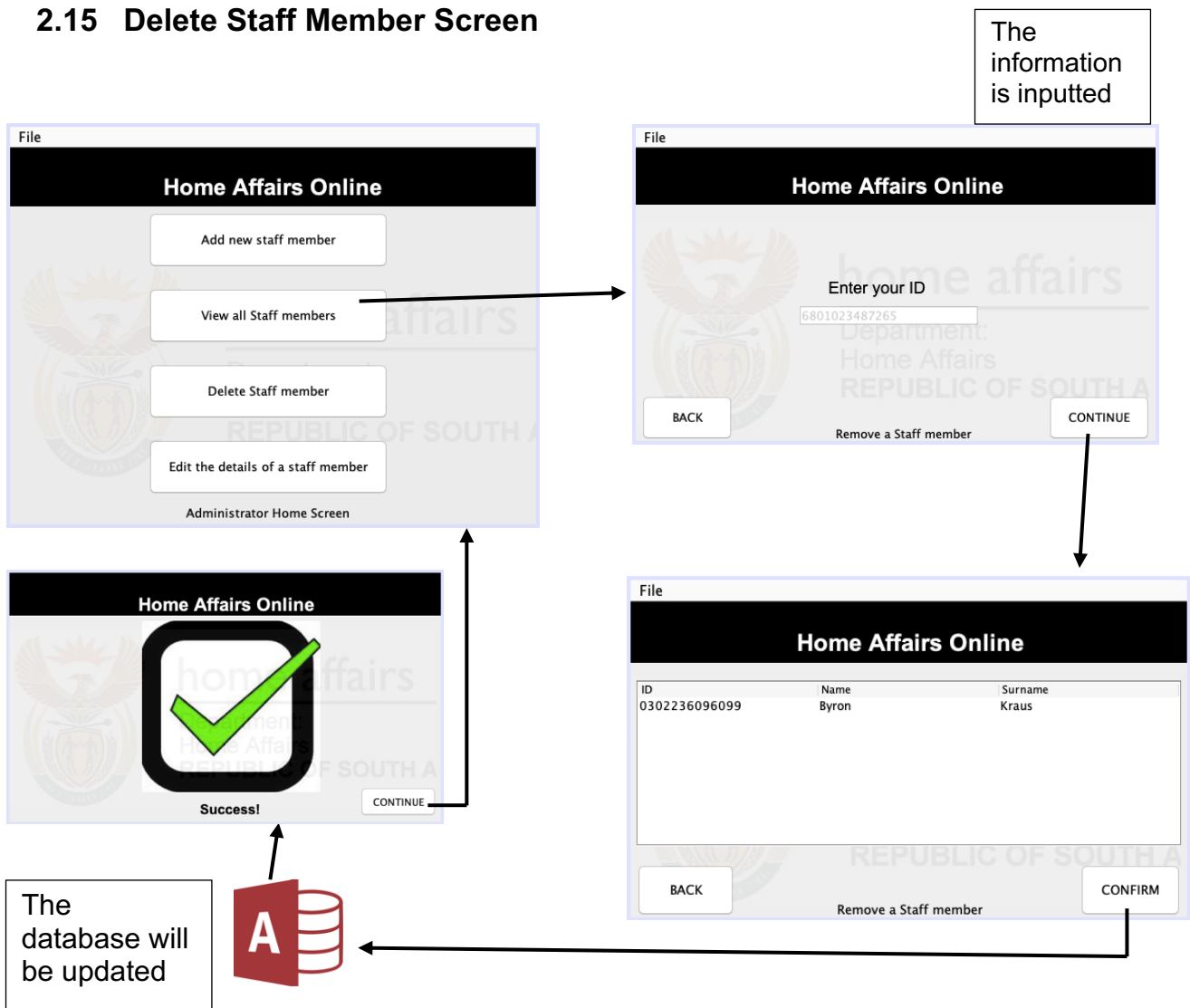
2.13 Add new Staff Member Screen



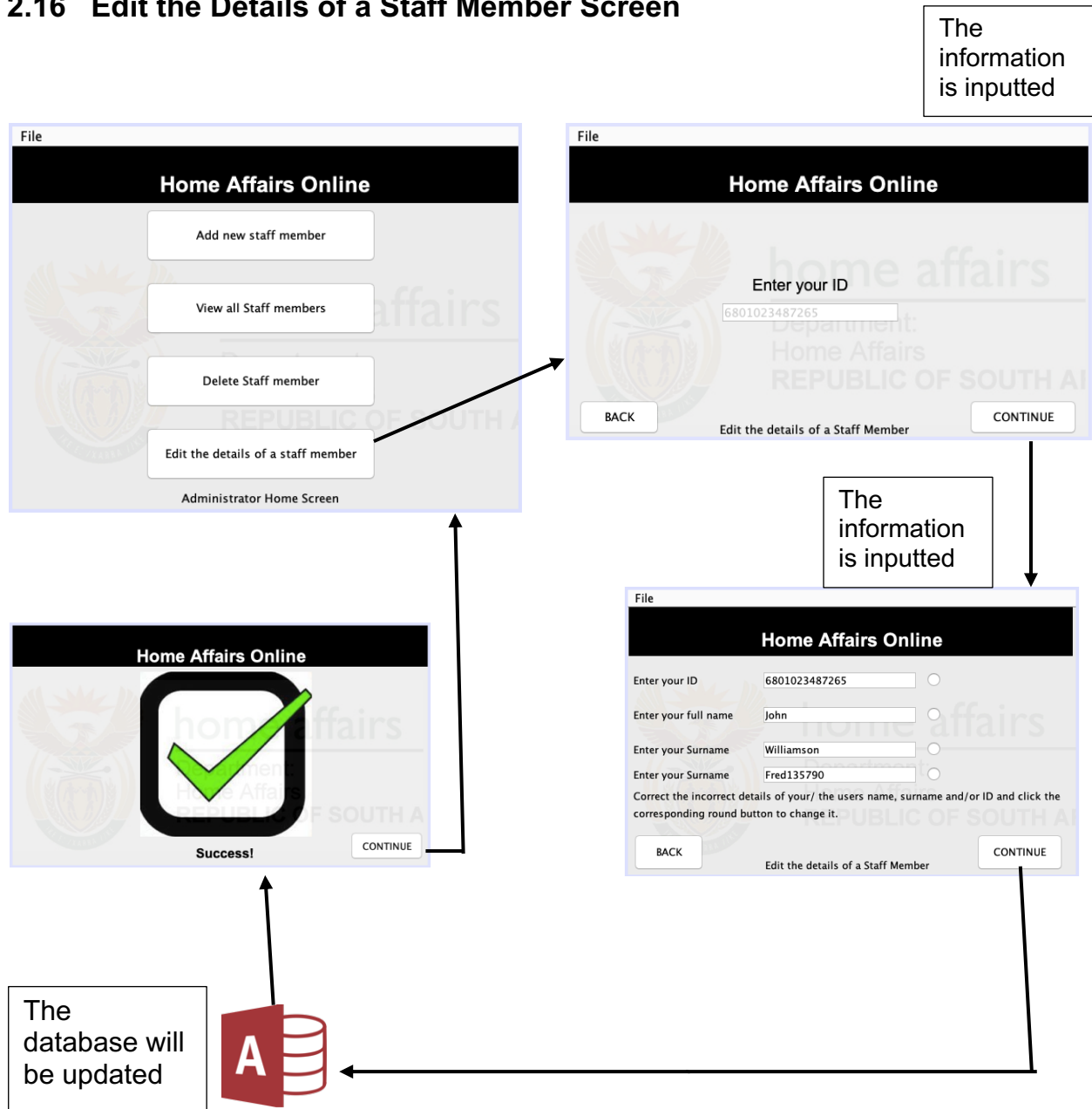
2.14 View all Staff Members Screen



2.15 Delete Staff Member Screen



2.16 Edit the Details of a Staff Member Screen



3 Class Design

Applicant	
Fields	
-String IDNumber	The ID Number of the applicant
-String firstName	The applicant's first name
-String middleName	The applicant's middle name
-String surname	The applicant's surname
-String dateOfBirth	The applicant's date of birth
-String cellNumber	The applicant's cellphone number
-String email	The applicant's email address
-String password	The applicant's password
Methods	
+Applicant(String cID, String fName, String mName, String sName, String DOB, String cNumber, String email, String pass)	Instantiates an object of the class and assigns values to each of the fields
+getID()	Returns the field IDNumber
+getFirstName()	Returns the field firstName
+getMiddleName()	Returns the field middleName
+getSurname()	Returns the field surname
+getDateOfBirth()	Returns the field dateOfBirth
+getCellNumber()	Returns the field cellNumber
+getEmail()	Returns the field email
+getPassword()	Returns the field password
+setID(String i)	Edits the field IDNumber
+setFirstName(String f)	Edits the field firstName
+setMiddleName(String m)	Edits the field middleName
+setSurname(String s)	Edits the field surname
+setCellNumber(String c)	Edits the field cellNumber
+setEmail(String e)	Edits the field email

ApplicantManager	
Fields	
-DatabaseManager dbm	This allows the link between the database and the program, and it allows for the manipulation of the tables
-Applicant [] applicantArr	This stores an array of Applicant objects
-int size	This stores the length of the array
Methods	
+ApplicantManager()	Instantiates an object of this class and fills the arrays with data using the database class
+addApplicant(String cID, String fName, String mName, String sName, String DOB, String cNumber, String email, String pass)	Takes the object's fields as a parameter and uses it to add the object as a record in the table
+getApplicant(String id) : Applicant	Takes the objects ID Number and uses it to return the corresponding object to the method
+deleteFromDatabase(String id)	Takes the applicants ID Number and uses it to delete the corresponding record in the table

Appointment	
Fields	
-Applicant applicant	A method from the Applicant class
-Category category	A method from the Category class
-String appointmentDateTime	The date and time of the appointment
Methods	
+Appointment(Applicant app, Category cat, String ADT)	Instantiates an object of the class and assigns values to each of the fields
+getApplicantID()	Returns the applicant's ID number in a user-friendly manner
+getCategoryName()	Returns the category name in a user-friendly manner
+getAppointmentDateTime()	Returns the field appointmentDateTime
+setAppointmentDateTime(String d)	Sets the field appointmentDateTime

Category	
Fields	
-String categoryID	A number which assigns the category name to a number
-String categoryName	The name of the category
Methods	
+Category(String cID, String cNM)	Instantiates an object of the class and assigns values to each of the fields
+getcategoryID()	Returns the categoryID
+getCategoryName()	Returns the categoryName.

UnabridgedBirthCertificate	
Fields	
-String applicationID	The unique number for the application
-Appointment appointmentID	An object from the Appointment class
-String StaffID	The unique number for the staff member
-BufferedImage parentIDNumber	The photo of the parents ID number
Methods	
+UnabridgedBirthCertificate(String aID, Appointment appID, String sID, BufferedImage pID)	Instantiates an object of the class and assigns values to each of the fields
+getApplicationID()	Returns the categoryID
+getAppointmentID()	Returns the appointmentID
+getStaffID()	Returns the staffID
+getParentIDNumber()	Returns the parentIDNumber
+setApplicationID(String a)	Edits the categoryID
+setAppointmentID(Appointment app)	Edits the appointmentID
+setStaffID(String s)	Edits the staffID
+setParentIDNumber(BufferedImage p)	Edits the parentIDNumber

UnabridgedBirthCertificateManager	
Fields	
-DatabaseManager dbm	This allows the link between the database and the program, and it allows for the manipulation of the tables
-UnabridgedBirthCertificate [] ubcArr	This stores an array of UnabridgedBirthCertificate objects
-int size	This stores the length of the array
Methods	
+UnabridgedBirthCertificateManager()	Instantiates an object of this class and fills the arrays with data using the database class
+addUnabridgedBirthCertificate(String aID, Appointment appID, String sID, BufferedImage pID)	Takes the object's fields as a parameter and uses it to add the object as a record in the table
+getUnabridgedBirthCertificate(String id) : UnabridgedBirthCertificate	Takes the objects ID Number and uses it to return the corresponding object to the method
+deleteFromDatabase(String id)	Takes the applicants ID Number and uses it to delete the corresponding record in the table
+getPastEmployeeHiredArray() : String[][]	This method retrieves a list of applicant objects from the database and selects the necessary fields to be displayed to the user. Then these details are stored in a 2d String array, which is returned.

Passport	
Fields	
-String applicationID	The unique number for the application
-Appointment appointmentID	An object from the Appointment class
-String StaffID	The unique number for the staff member
-BufferedImage IDNumber	The photo of the applicants ID number
Methods	
+Passport(String aID, Appointment appID, String sID, BufferedImage pID)	Instantiates an object of the class and assigns values to each of the fields
+getApplicationID()	Returns the categoryID
+getAppointmentID()	Returns the appointmentID
+getStaffID()	Returns the staffID
+getIDNumber()	Returns the IDNumber
+setApplicationID(String a)	Edits the categoryID
+setAppointmentID(String app)	Edits the appointmentID
+setStaffID(String s)	Edits the staffID
+setIDNumber(BufferedImage i)	Edits the IDNumber

PassportManager	
Fields	
-DatabaseManager dbm	This allows the link between the database and the program, and it allows for the manipulation of the tables
-Passport [] pArr	This stores an array of Passport objects
-int size	This stores the length of the array
Methods	
+Passport()	Instantiates an object of this class and fills the arrays with data using the database class
+addPassport(String aID, Appointment appID, String sID, BufferedImage pID)	Takes the object's fields as a parameter and uses it to add the object as a record in the table
+getPassport(String id) : Passport	Takes the objects ID Number and uses it to return the corresponding object to the method
+deleteFromDatabase(String id)	Takes the applicants ID Number and uses it to delete the corresponding record in the table
+getPastPassport() : String[][]	This method retrieves a list of Applicant objects from the database and selects the necessary fields to be displayed to the user. Then these details are stored in a 2d String array, which is returned.

SmartID	
Fields	
-String applicationID	The unique number for the application
-Appointment appointmentID	An object from the Appointment class
-String StaffID	The unique number for the staff member
-BufferedImage passport	The photo of the applicants passport
-BufferedImage parentIDNumber	The photo of the applicants parents ID number
-BufferedImage birthCertificate	The photo of the applicants birth certificate
-BufferedImage proofOfResidence	The photo of the applicants proof of residence
Methods	

+SmartID(String aID, Appointment appID, String sID, BufferedImage p, BufferedImage pID, BufferedImage b, BufferedImage POR)	Instantiates an object of the class and assigns values to each of the fields
+getApplicationID()	Returns the categoryID
+getAppointmentID()	Returns the appointmentID
+getStaffID()	Returns the staffID
+getPassport()	Returns the passport
+getParentIDNumber()	Returns the parentIDNumber
+getBirthCertificate()	Returns the birthCertificate
+getIDNumber()	Returns the IDNumber
+setApplicationID(String a)	Edits the categoryID
+setAppointmentID(String app)	Edits the appointmentID
+setStaffID(String s)	Edits the staffID
+setPassport(BufferedImage p)	Edits the passport
+setParentIDNumber(BufferedImage i)	Edits the parentIDNumber
setBirthCertificate(BufferedImage b)	Edits the birthCertificate
+setProofOfResidence(BufferedImage POR)	Edits the proofOfResidence

SmartIDManager	
Fields	
-DatabaseManager dbm	This allows the link between the database and the program, and it allows for the manipulation of the tables
-SmartID [] sArr	This stores an array of SmartID objects
-int size	This stores the length of the array
Methods	
+SmartID()	Instantiates an object of this class and fills the arrays with data using the database class
+addSmartID(String aID, Appointment appID, String sID, BufferedImage p, BufferedImage pID, BufferedImage b, BufferedImage POR)	Takes the object's fields as a parameter and uses it to add the object as a record in the table
+getSmartID(String id) : SmartID	Takes the objects ID Number and uses it to return the corresponding object to the method
+deleteFromDatabase(String id)	Takes the applicants ID Number and uses it to delete the corresponding record in the table

+ getPastEmployeeHiredArray() : String[][]	This method retrieves a list of Applicant objects from the database and selects the necessary fields to be displayed to the user. Then these details are stored in a 2d String array, which is returned.
--	--

Staff	
Fields	
-String staffIDNumber	The ID Number of the staff member
-String roleID	The staff members role
-String firstName	The applicant's first name
-String middleName	The staff members middle name
-String surname	The staff members surname
-String password	The applicant's password
Methods	
+Staff(String sID, String rID, String fName, String mName, String sName, String pass)	Instantiates an object of the class and assigns values to each of the fields
+getID()	Returns the field staffIDNumber
+getRoleID()	Returns the field roleID
+getFirstName()	Returns the field firstName
+getMiddleName()	Returns the field middleName
+getSurname()	Returns the field surname
+getPassword()	Returns the field password
+setID(String i)	Edits the field IDNumber
+setFirstFirstName(String f)	Edits the field firstName
+setMiddleName(String m)	Edits the field middleName
+setSurname(String s)	Edits the field surname

StaffManager	
Fields	
-DatabaseManager dbm	This allows the link between the database and the program, and it allows for the manipulation of the tables
-Staff [] staffArr	This stores an array of Staff objects
-int size	This stores the length of the array
Methods	
+StaffManager()	Instantiates an object of this class and fills the arrays with data using the database class

+addStaff(String sID, String rID, String fName, String mName, String sName, String pass)	Takes the object's fields as a parameter and uses it to add the object as a record in the table
+getStaff(String id) : Staff	Takes the objects ID Number and uses it to return the corresponding object to the method
+deleteFromDatabase(String id)	Takes the staff members ID Number and uses it to delete the corresponding record in the table
+getEmployeeHiredArray() : String[][]	This method retrieves a list of Employee objects from the database and selects the necessary fields to be displayed to the user. Then these details are stored in a 2d String array, which is returned.

Role	
Fields	
-String roleID	A number which assigns the role name to a number
-String roleDescription	The name of the role of the staff member
Methods	
+Role(String rID, String rD)	Instantiates an object of the class and assigns values to each of the fields
+getRoleID()	Returns the roleID
+getRoleDescription()	Returns the roleDescription.

4 Persistent Storage Design

tblApplicant

tblApplicant	tblAppointment	tblCategory	tblPassport	tblRole	tblSmartID	tbl
Field Name	Data Type	Description				
applicant_ID	AutoNumber	This is a primary key and is an autonumber for the applicants				
firstname	Short Text	This is the firstname of the applicant				
middlename	Short Text	This field is the middlename of the applicant				
surname	Short Text	This is the surname of the applicant				
idnumber	Number	This field is the ID number of the applicant				
dateofbirth	Short Text	This is the date of birth of the applicant				
cellno	Short Text	This field stores the cell phone number of the applicant				
email	Short Text	This is the email address of the applicant				
password	Short Text	This field stores the password of the applicant				

tblCategory	tblPassport	tblRole	tblSmartID	tblStaff	tblUnabridgedBirthCertificate	tblApplicant		
applicant_id	firstname	middlename	surname	idnumber	dateofbirth	cellno	email	password
1	Lionel	Andre	Messi	7505256520082	25/05/1975	0828525405	Lmessi@gmail.com	MessiTheGoat1975
2	Cristiano		Ronaldo	8101070780083	07/01/1981	0832205553	Cronaldo@icloud.com	CR7forTheGold
3	Xavier	Hernandes	Creus	7710080382085	08/10/1977	0844899578	XavierTheKnight@gmail.com	Xavi47Creus
4	Zlatan		Ibrahimovic	6507125553089	12/07/1965	0836146497	ZlatanI@gmail.com	Zlatan1STank
5	Neymar	da Silva	Santos Jr	6510185051085	18/10/1965	0844015301	NeymarJR@icloud.com	NeymarJR296
*	(New)			0				

tblAppointment

tblAppointment	tblCategory	tblPassport	tblRole	tblSmartID
appointment_ID	applicant_ID	category_ID	appointmer	Click to Add
1	1	1	12/01/2020 15:	
2	4	3	13/01/2020 09:	
3	2	2	13/01/2020 11:	
*	(New)			

tblAppointment	tblCategory	tblPassport	tblRole	tblSmartID	tblStaff	tblUnabridgedBirthCertificate
Field Name	Data Type	Description (Optional)				
appointment_ID	AutoNumber	This is a primary key that creates a unique number for the appointments				
applicant_ID	Number	This is a foreign key and stores the unique number of the applicant				
category_ID	Number	This is a foreign key and stores the numbers 1, 2 or 3 to confirm the type of staff member they are				
appointmentdatetime	Short Text	This stores the date and time of the appointment of the applicants				

tblCategory

tblCategory	tblPassport
category_ID	categoryname
1	UnabridgedBirthCertificate
2	Passport
3	SmartID
*	

Field Name	Data Type	Description (Optional)
category_ID	Number	This is the primary key and creates the link with tblAppointment with numbers 1, 2 or 3
categoryname	Short Text	This stores the names of the type of booking that the applicant has booked for

tblUnabridgedBirthCertificate

tblUnabridgedBirthCertificate	tblAppointment	tblStaff
application_ID	appointmentmer	staff_ID
1	1	0(0)
2	2	0(0)
(New)		0(0)

Field Name	Data Type	Description (Optional)
application_ID	AutoNumber	This is a primary key and stores the unique number for the application
appointment_ID	Number	This is a foreign key and has the appointment number
staff_ID	Number	This is a foreign key and stores the numbr for the staff member which helped assit the applicant in booking
parentidnumber	Attachment	This is an attachment that stores the photo of the parents ID

tblPassport

tblPassport	tblRole	tblSmartID	tblStaff
application_ID	appointmentmer	staff_ID	
1	3	1	0(0)
2	1	2	0(0)
3	4		0(0)
(New)			0(0)

Field Name	Data Type	Description (Optional)
application_ID	AutoNumber	This is a primary key and stores the unique number for the application
appointment_ID	Number	This is a foreign key and has the appointment number
staff_ID	Number	This is a foreign key and stores the numbr for the staff member which helped assit the applicant in booking
idnumber	Attachment	This is an attachment and stores the photo of the ID of the applicant

tblSmartID

tblSmartID	tblStaff	tblUnabridgedBirthCertificate				
application_ID	appointmentmer	staff_ID				
1	4		0(0)	0(0)	0(0)	0(0)
2	5	1	0(0)	0(0)	0(0)	0(0)
3	6	2	0(0)	0(0)	0(0)	0(0)
*(New)			0(0)	0(0)	0(0)	0(0)

tblSmartID	tblStaff	tblUnabridgedBirthCertificate	
Field Name	Data Type	Description (Optional)	
application_ID	AutoNumber	This is a primary key and stores the unique number for the application	
appointment_ID	Number	This is a foreign key and has the appointment number	
staff_ID	Number	This is a foreign key and stores the numbr for the staff member which helped assit the applicant in booking	
passport	Attachment	This is an attachment that stores the photo of the applicants passport	
parentidnumber	Attachment	This is an attachment that stores the photo of the parents ID	
birthcertificate	Attachment	This is an attachment that stores the photo of the applicants birth certificate	
proofofresidence	Attachment	This is an attachment that stores the photo of the proof of residence of the applicant	

tblStaff

tblStaff	tblUnabridgedBirthCertificate					
staff_ID	role_id	firstname	middlenam	surname	password	C
1	1	David		Knowles	Dknowles422	
2	1	Hubert	Gerry	Goedeke	HGG4theWin	
3	2	Geoff		Thomas	ThomasG1293	
4	2	Peter		Farquharson	Farq1327Peter	
5	2	Fiona		King	Fiona4King	
6	2	Carol	De	Matteis	Carol4De8Mat	
*(New)						

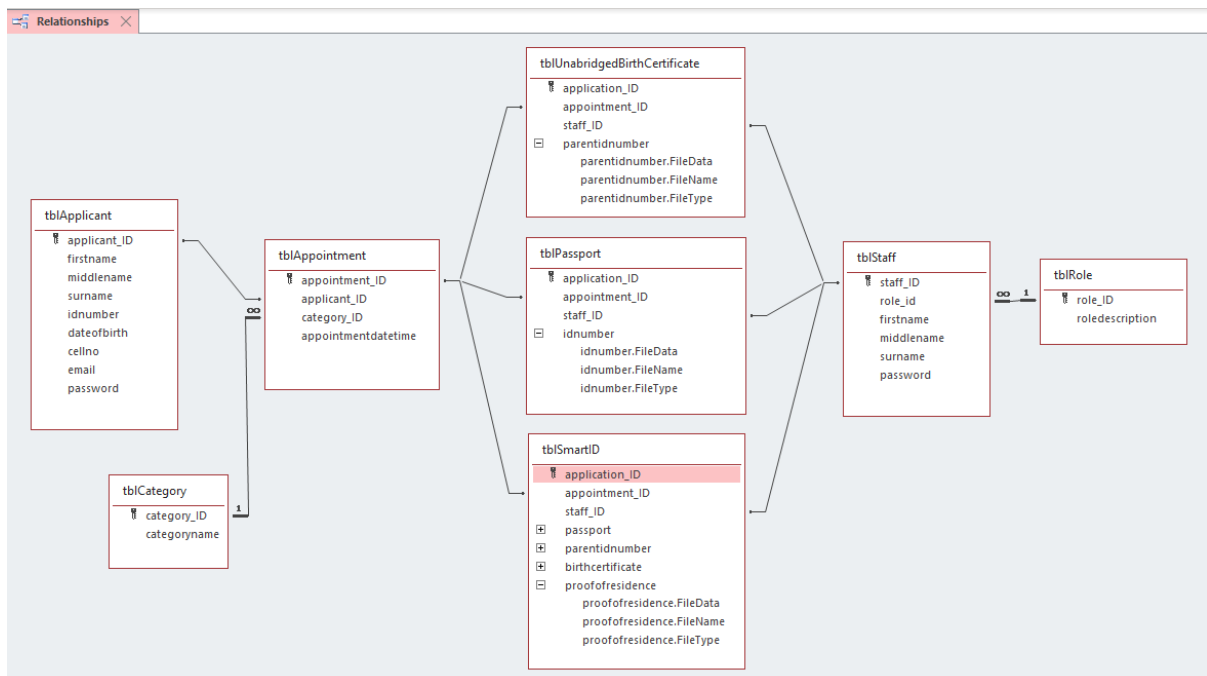
tblStaff	tblUnabridgedBirthCertificate	
Field Name	Data Type	Description (Optional)
staff_ID	AutoNumber	This is the primary key and stores a unique number for the staff
role_id	Number	This is a foreign key and link to tblRole where the role of the staff member is
firstname	Short Text	This is the firstname of the staff member
middlename	Short Text	This is the middle name of the staff member
surname	Short Text	This is the surname of the staff member
password	Short Text	This is the password of the staff member

tblRole

tblRole	tblSmartID
role_ID	roledescription
	Administrator
	2 Staff Member
*	

tblRole	tblSmartID	tblStaff	tblUnabridgedBirthCertificate
Field Name	Data Type	Description (Optional)	
role_ID	Number	This is the primary key and differentiates between staff members and administrators	
roledescription	Short Text	This links the numbers 1 and 2 to staff members and administrators	

Relationship table



5 Explanation of Storage Design

I have used tables in a database made in MS Access to store majority of my data. I have normalized my data into third normal form to insure that I avoid any of the common anomalies (such as an insert, delete and/or update anomaly). I have also kept data redundancy to a minimum. I decided to use MS Access as the data is more secure in a database, also, using SQL statements I can manipulate the data within the necessary tables. This is also far more safe then other methods of storage such as text files and text files have a very complex method for manipulating data. MS Access also allows all the data to be stored in one place instead of having to access and search through multiple different text files. My database has inter-field relationships that connect the tables together which makes the SQL statement simpler to execute to manipulate the data. I can also locate the data in other tables, to be shown or manipulated when data is given to me and when that data is from another table. Text files are incapable of doing the above capabilities, which causes the table to be far more effective and efficient to use.

All usernames (the username is the ID of the applicant or staff member) and passwords of the employees, staff members and administrators are stored in the database because I find it better manipulating data in databases. Text files are also unreliable data can be lost or become corrupt more easily. The login details are safe. Also, if there is an error in the database, users will be unable to login to the system. But if they were able to login without the login details, the home affairs program could be susceptible to SQL injection attacks.

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